

1. Basic Server Controls

Source Code (Form1):

```
using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Drawing;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
using System.Windows.Forms;

namespace Basic_Server_Controls
{
    public partial class Form1 : Form
    {
        public Form1()
        {
            InitializeComponent();
        }

        private void button1_Click(object sender, EventArgs e)
        {
            if (textBox1.Text == "")
            {
                MessageBox.Show("Enter Username !!");
            }
            else if (textBox2.Text == "")
            {
                MessageBox.Show("Enter Password !!");
            }
            else if (textBox1.Text == "Abi" && textBox2.Text == "123")
            {
                this.Hide();
                Form frm2 = new Form2();
                frm2.ShowDialog();
            }
            else
            {
                MessageBox.Show("Invalid Credential", "Try Again");
            }
        }
    }
}
```

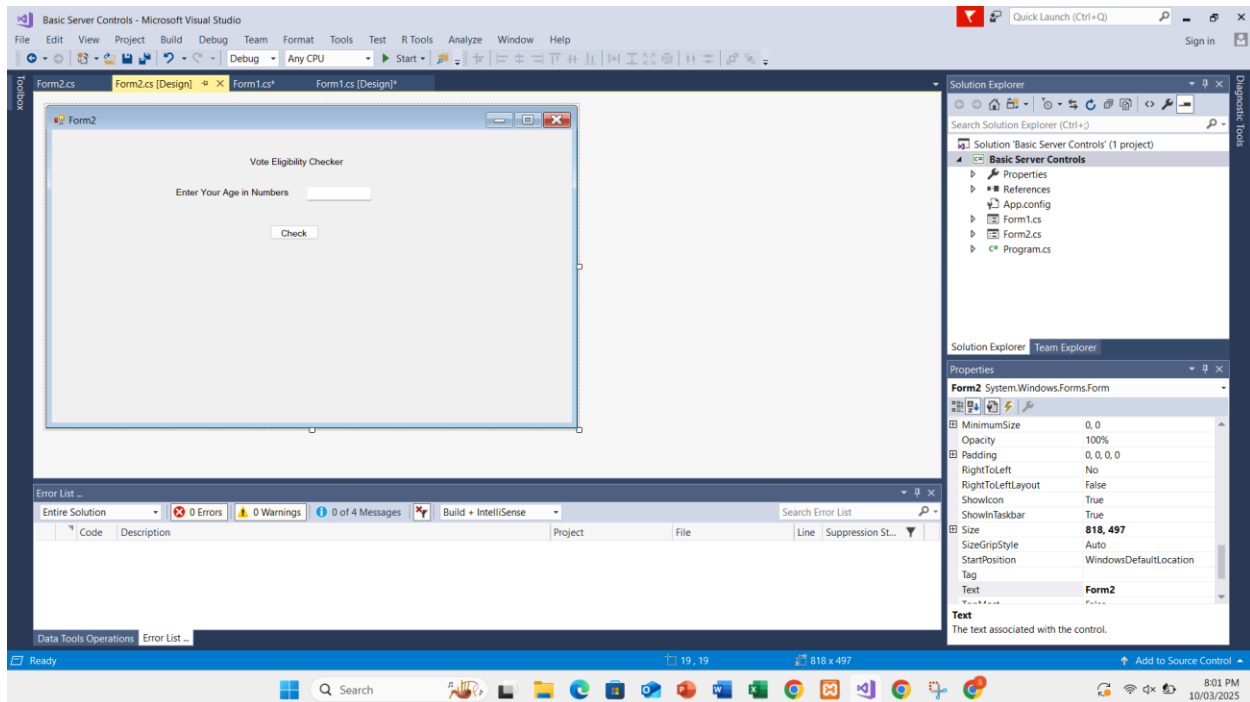
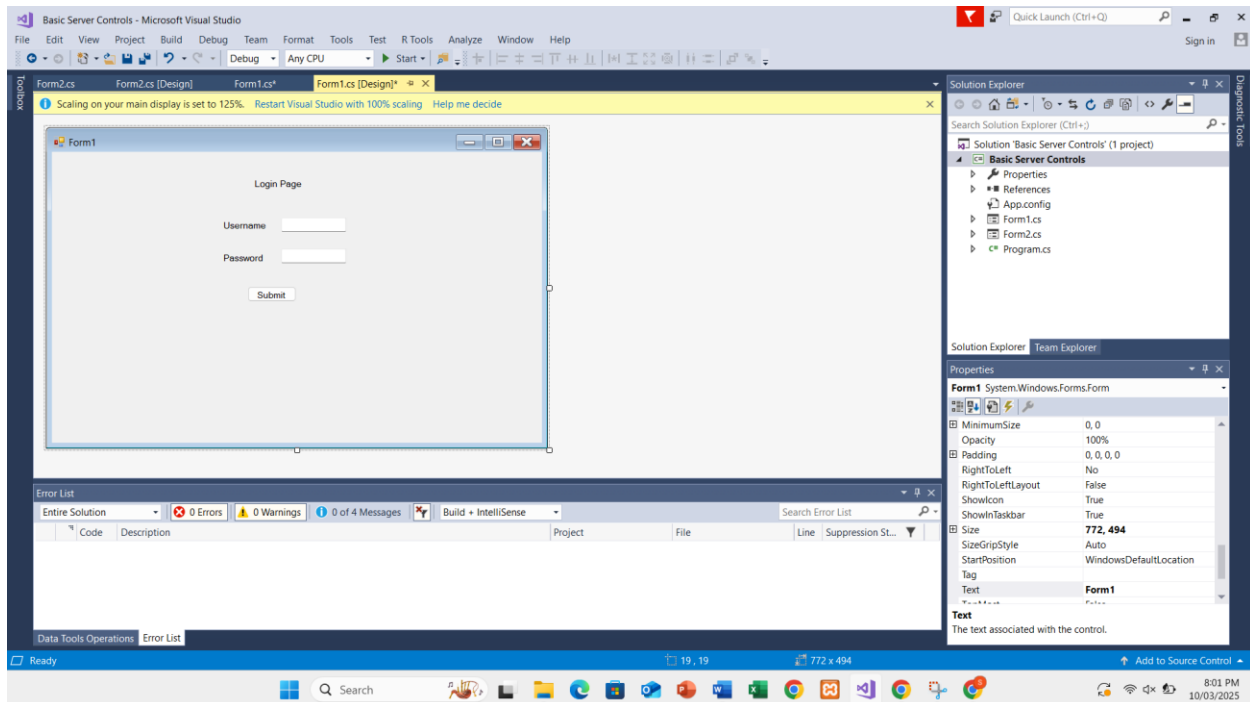
Form2:

```
using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Drawing;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
using System.Windows.Forms;

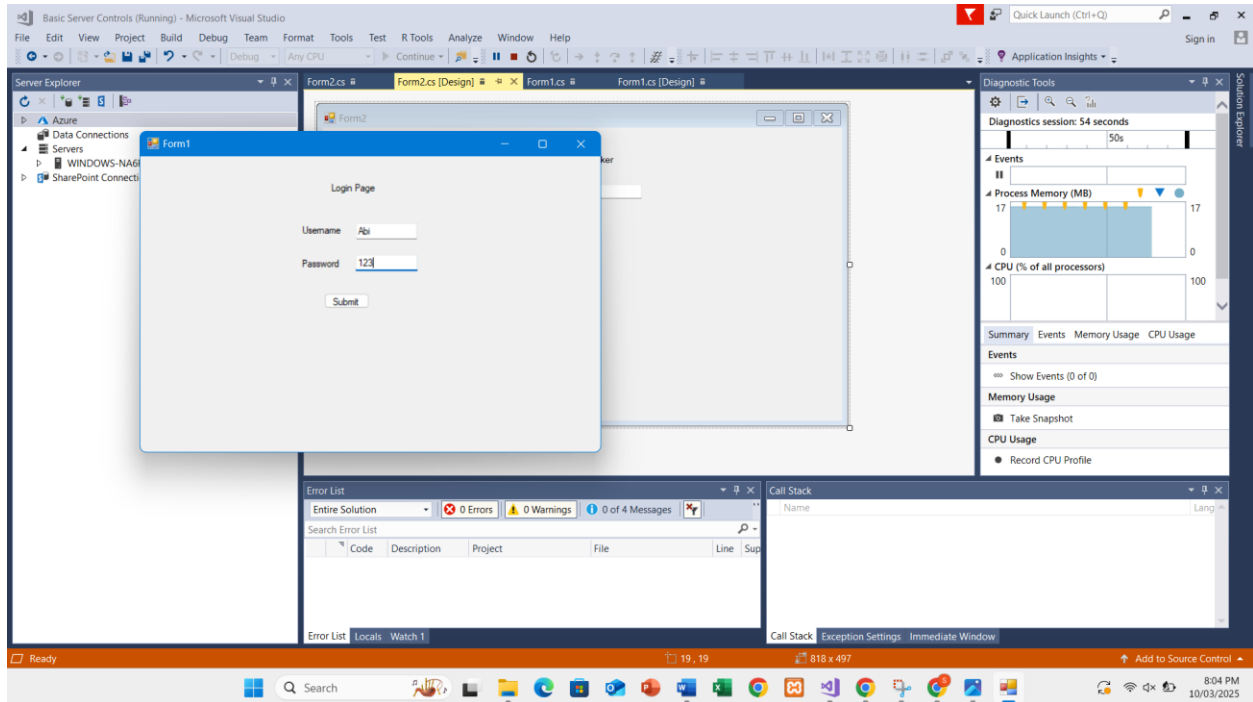
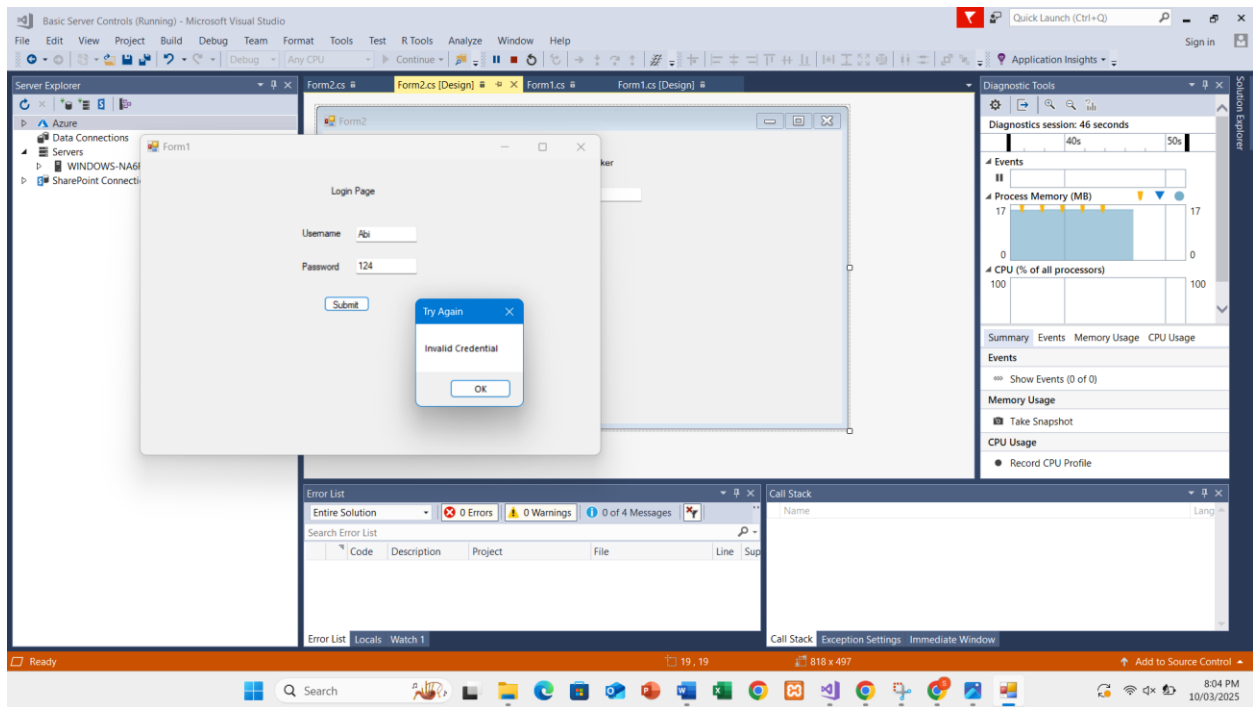
namespace Basic_Server_Controls
{
    public partial class Form2 : Form
    {
        public Form2()
        {
            InitializeComponent();
        }

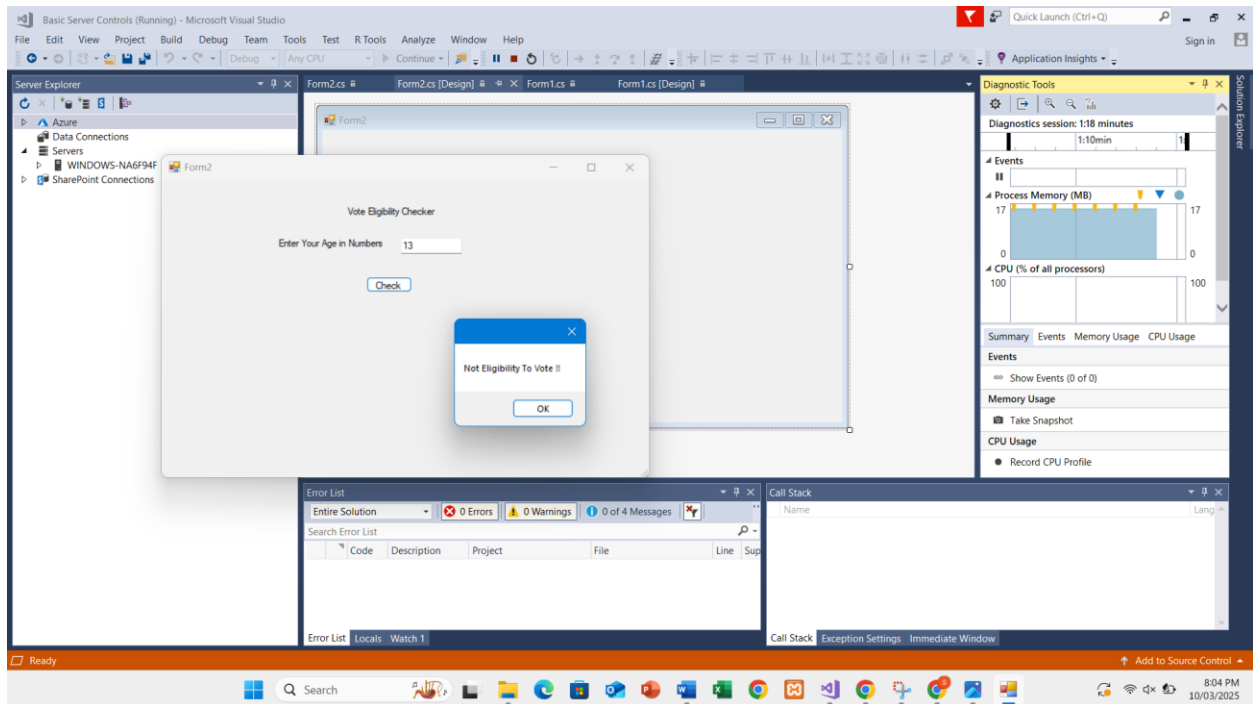
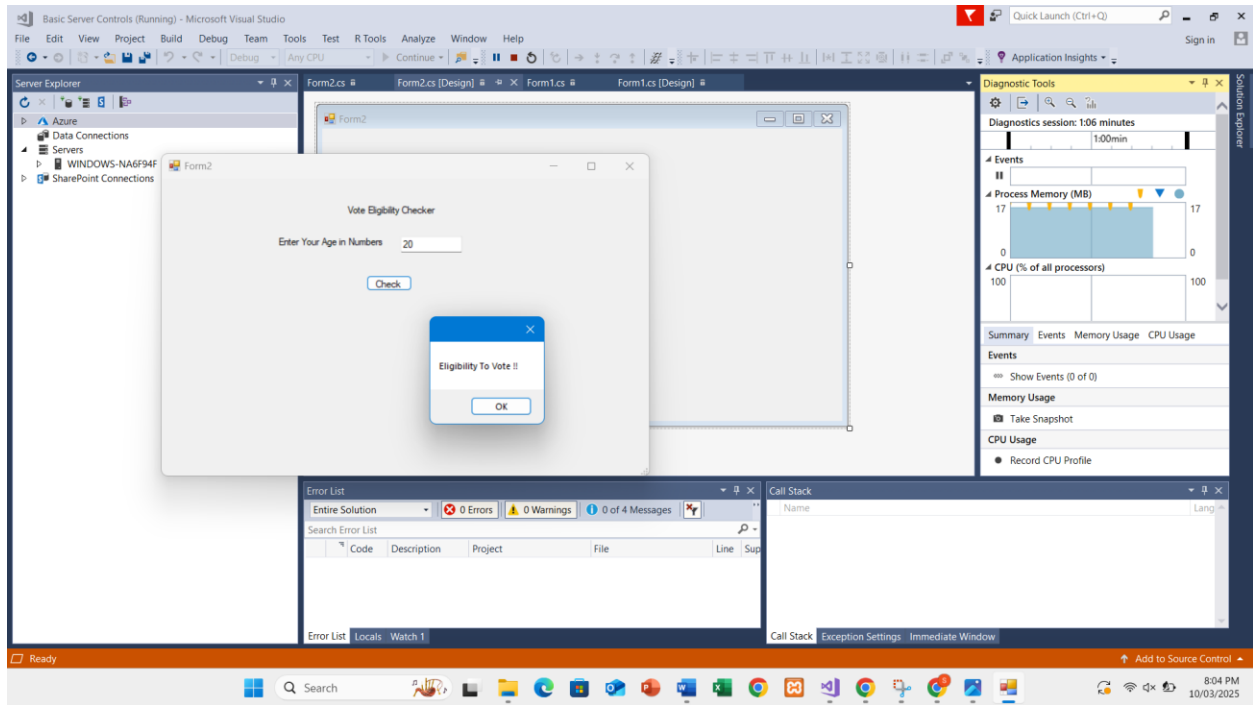
        private void button1_Click(object sender, EventArgs e)
        {
            int a;
            a = Convert.ToInt32(textBox1.Text);
            if (a >= 18)
            {
                MessageBox.Show("Eligibility To Vote !!");
            }
            else
            {
                MessageBox.Show("Not Eligibility To Vote !!");
            }
        }
    }
}
```

Design Page:



Output:





2. Validation Class

Source Code:

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="WebForm1.aspx.cs"
Inherits="Validation_Class.WebForm1" %>

<!DOCTYPE html>

<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
    <title></title>
    <style type="text/css">
        .auto-style1 {
            width: 308px;
        }
    </style>
</head>
<body>
    <form id="form1" runat="server">
        <div>
            <table style="width:100%;">
                <tr>
                    <td class="auto-style1">Registration Form</td>
                    <td>&nbsp;</td>
                    <td>&nbsp;</td>
                </tr>
                <tr>
                    <td class="auto-style1">
                        <asp:Label ID="Label1" runat="server" Text="Username"></asp:Label>
                    </td>
                    <td>
                        <asp:TextBox ID="TextBox1" runat="server"
OnTextChanged="TextBox1_TextChanged"></asp:TextBox>
                        <asp:RequiredFieldValidator ID="RequiredFieldValidator1" runat="server"
ControlToValidate="TextBox1" ErrorMessage="Enter valid
Username"></asp:RequiredFieldValidator>
                    </td>
                    <td>&nbsp;</td>
                </tr>
                <tr>
                    <td class="auto-style1">
                        <asp:Label ID="Label2" runat="server" Text="Age"></asp:Label>
                    </td>
                    <td>
                        <asp:TextBox ID="TextBox2" runat="server"
OnTextChanged="TextBox1_TextChanged"></asp:TextBox>
                    </td>
                    <td>&nbsp;</td>
                </tr>
            </table>
        </div>
    </form>
</body>
</html>
```

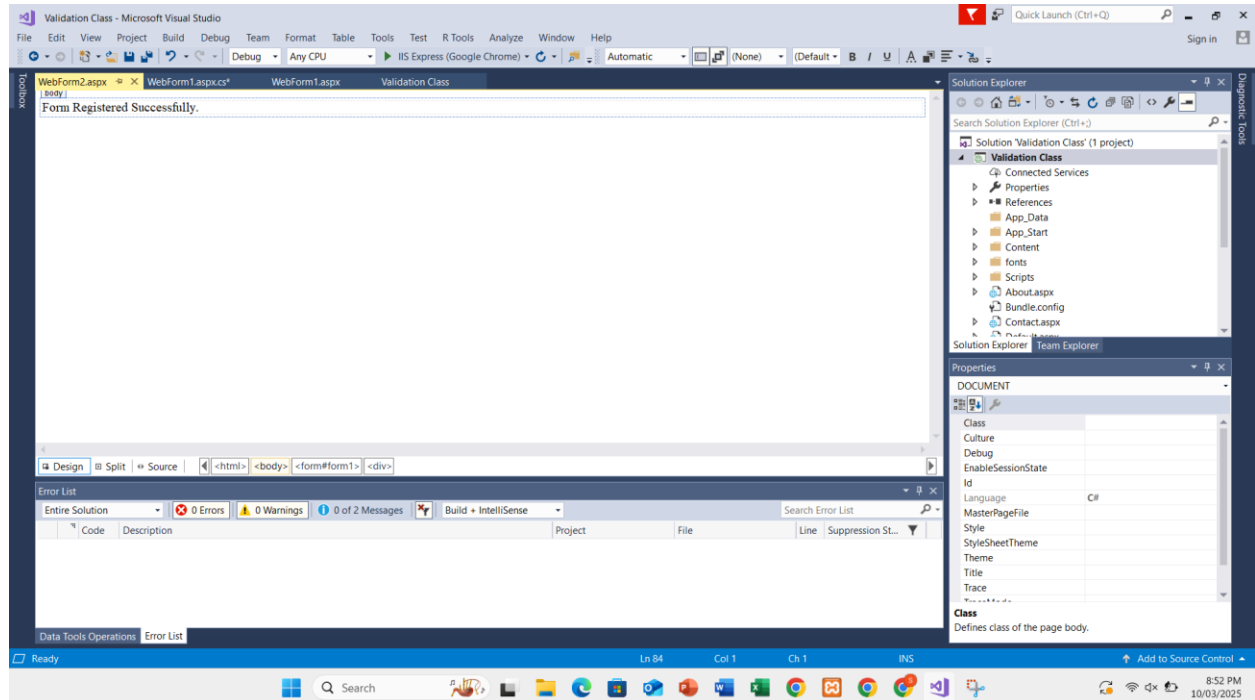
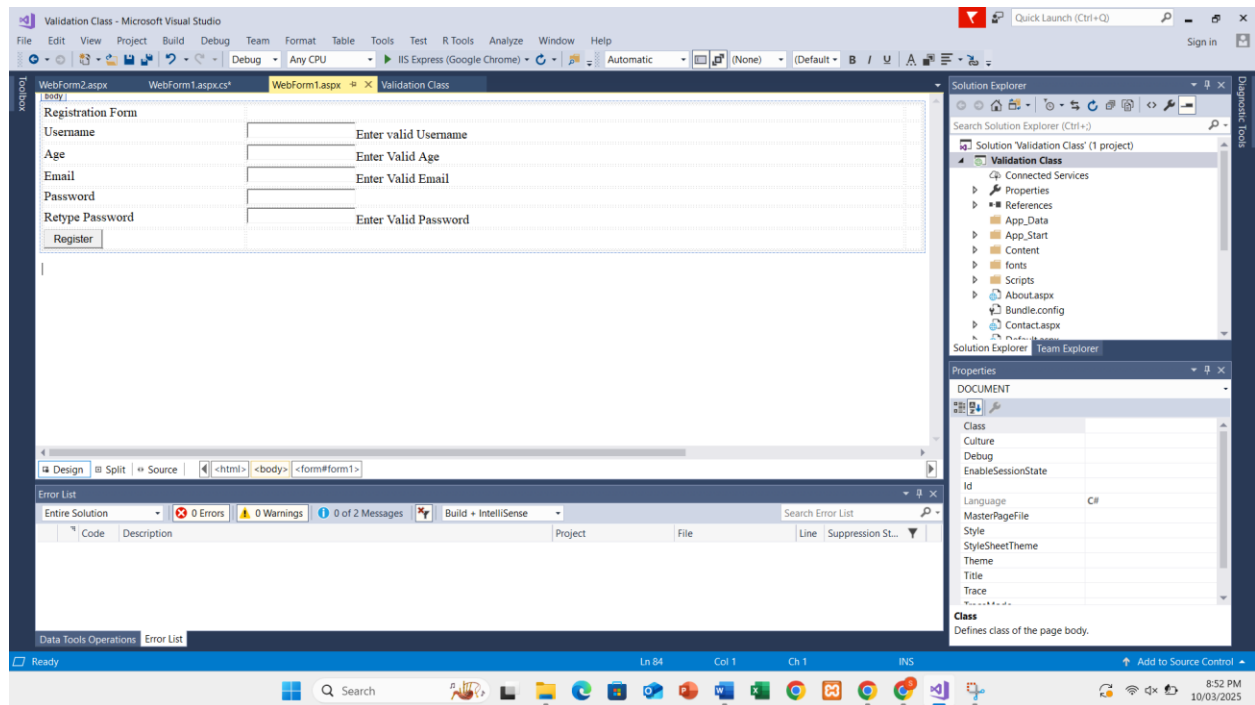
```

        <asp:RangeValidator ID="RangeValidator1" runat="server"
ControlToValidate="TextBox2" ErrorMessage="Enter Valid Age" MaximumValue="50"
MinimumValue="30"></asp:RangeValidator>
    </td>
    <td>&nbsp;</td>
</tr>
<tr>
    <td class="auto-style1">
        <asp:Label ID="Label3" runat="server" Text="Email"></asp:Label>
    </td>
    <td>
        <asp:TextBox ID="TextBox3" runat="server"
OnTextChanged="TextBox1_TextChanged"></asp:TextBox>
        <asp:RegularExpressionValidator ID="RegularExpressionValidator1"
runat="server" ControlToValidate="TextBox3" ErrorMessage="Enter Valid Email"
ValidationExpression="\w+([-+.']\w+)*@\w+([-.\]\w+)*\.\w+([-
.\]\w+)*"></asp:RegularExpressionValidator>
    </td>
    <td>&nbsp;</td>
</tr>
<tr>
    <td class="auto-style1">
        <asp:Label ID="Label4" runat="server" Text="Password"></asp:Label>
    </td>
    <td>
        <asp:TextBox ID="TextBox4" runat="server"
OnTextChanged="TextBox1_TextChanged" Type="Password"></asp:TextBox>
    </td>
    <td>&nbsp;</td>
</tr>
<tr>
    <td class="auto-style1">
        <asp:Label ID="Label5" runat="server" Text="Retype Password"></asp:Label>
    </td>
    <td>
        <asp:TextBox ID="TextBox5" runat="server"
OnTextChanged="TextBox1_TextChanged" Type="Password"></asp:TextBox>
        <asp:CompareValidator ID="CompareValidator1" runat="server"
ControlToCompare="TextBox4" ControlToValidate="TextBox5" ErrorMessage="Enter Valid
Password"></asp:CompareValidator>
    </td>
    <td>&nbsp;</td>
</tr>
<tr>
    <td class="auto-style1">

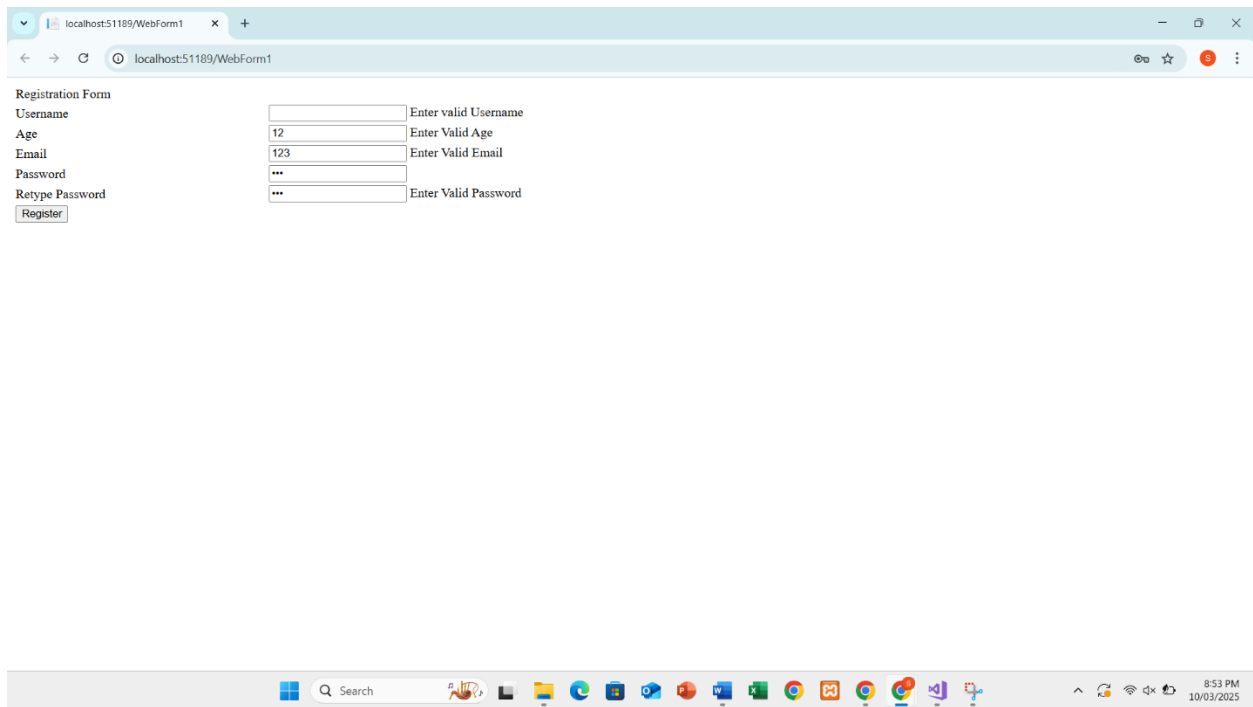
```

```
        <asp:Button ID="Button1" runat="server" Text="Register"
OnClick="Button1_Click" PostBackUrl="~/WebForm2.aspx" />
    </td>
    <td>&nbsp;</td>
    <td>&nbsp;</td>
</tr>
</table>
</div>
</form>
</body>
</html>
```


Design Page:



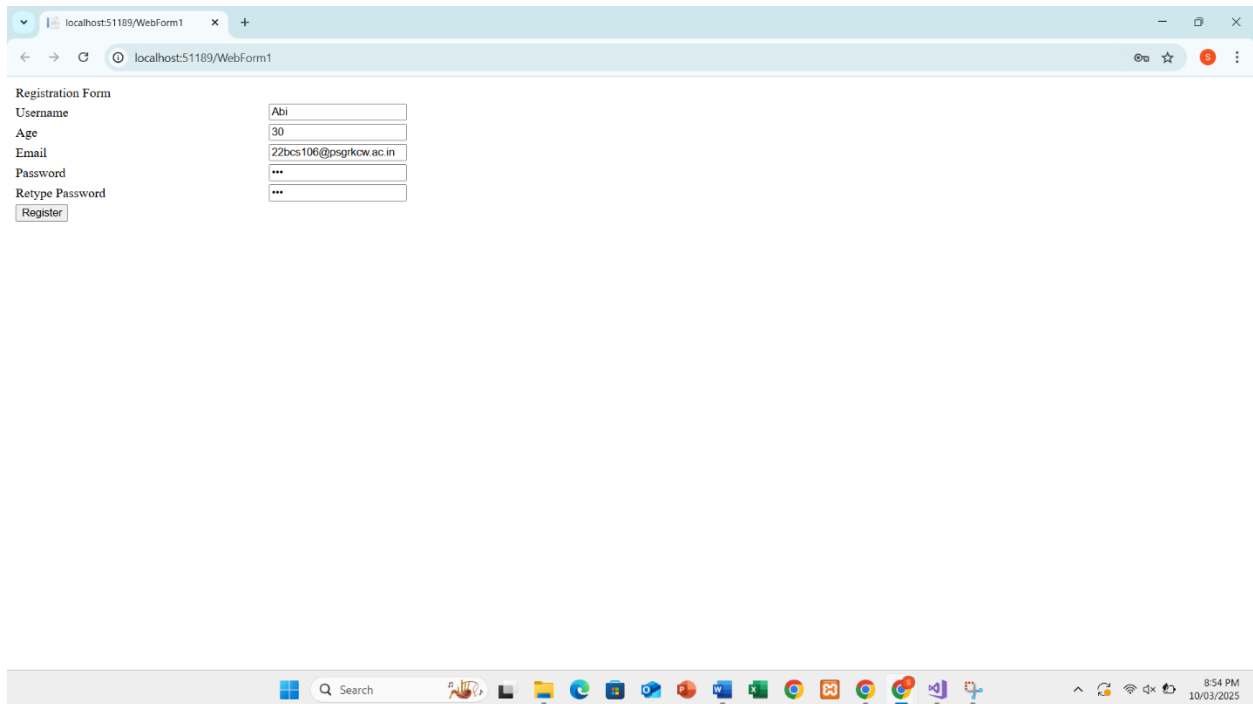
Output:



A screenshot of a web browser window displaying a registration form. The browser's address bar shows 'localhost:51189/WebForm1'. The form is titled 'Registration Form' and contains five input fields with corresponding validation messages:

Field	Validation Message
Username	Enter valid Username
Age	Enter Valid Age
Email	Enter Valid Email
Password	
Retype Password	Enter Valid Password

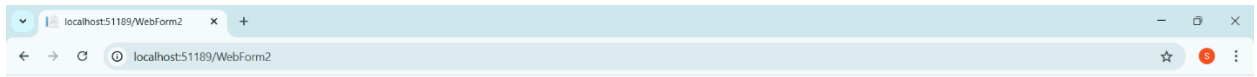
Below the fields is a 'Register' button. The Windows taskbar at the bottom shows the time as 8:53 PM on 10/03/2025.



A second screenshot of the same web browser window, showing the registration form after user input. The 'Register' button is now disabled. The input fields contain the following data:

Field	Value
Username	Abi
Age	30
Email	22bcs106@psgkwc.ac.in
Password	***
Retype Password	***

The 'Register' button is disabled. The Windows taskbar at the bottom shows the time as 8:54 PM on 10/03/2025.



Form Registered Successfully.



3. Iterative Structures

Source Code:

```
using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Drawing;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
using System.Windows.Forms;

namespace Student_Mark
{
    public partial class Form1 : Form
    {
        string[] StudentNames = new string[100];
        int[,] StudentMarks = new int[100, 3];
        int studentcount = 0;
        public Form1()
        {
            InitializeComponent();
        }

        private void Add_Click(object sender, EventArgs e)
        {
            if (studentcount <= 100)
            {
                MessageBox.Show("student list full");
            }

            StudentNames[studentcount] = textBox1.Text;
            StudentMarks[studentcount, 0] = Convert.ToInt32(textBox2.Text);
            StudentMarks[studentcount, 1] = Convert.ToInt32(textBox3.Text);
            StudentMarks[studentcount, 2] = Convert.ToInt32(textBox4.Text);
            studentcount++;

            MessageBox.Show("student added successfully");
            ClearInputs();
        }

        private void Diaplay_Click(object sender, EventArgs e)
        {

```

```

        listBox1.Items.Clear();
        for (int i = 0; i < studentcount; i++)
        {
            listBox1.Items.Add(StudentNames[i]+"marks:"+StudentMarks[i,
0]+"\n"+StudentMarks[i, 1]+"\n"+StudentMarks[i, 2]);
        }
    }

    private void Average_Click(object sender, EventArgs e)
    {
        if (studentcount == 0)
        {
            MessageBox.Show(" No students available ");
            return;
        }

        double totalmarks = 0;
        int totalsubjets = studentcount * 3;
        for (int i = 0; i < studentcount; i++)
        {
            for (int j = 0; j < 3; j++)
            {
                totalmarks += StudentMarks[i, j];
            }
        }
        double average = totalmarks / totalsubjets;
        MessageBox.Show($"class Average Marks: {average:F2}");
    }

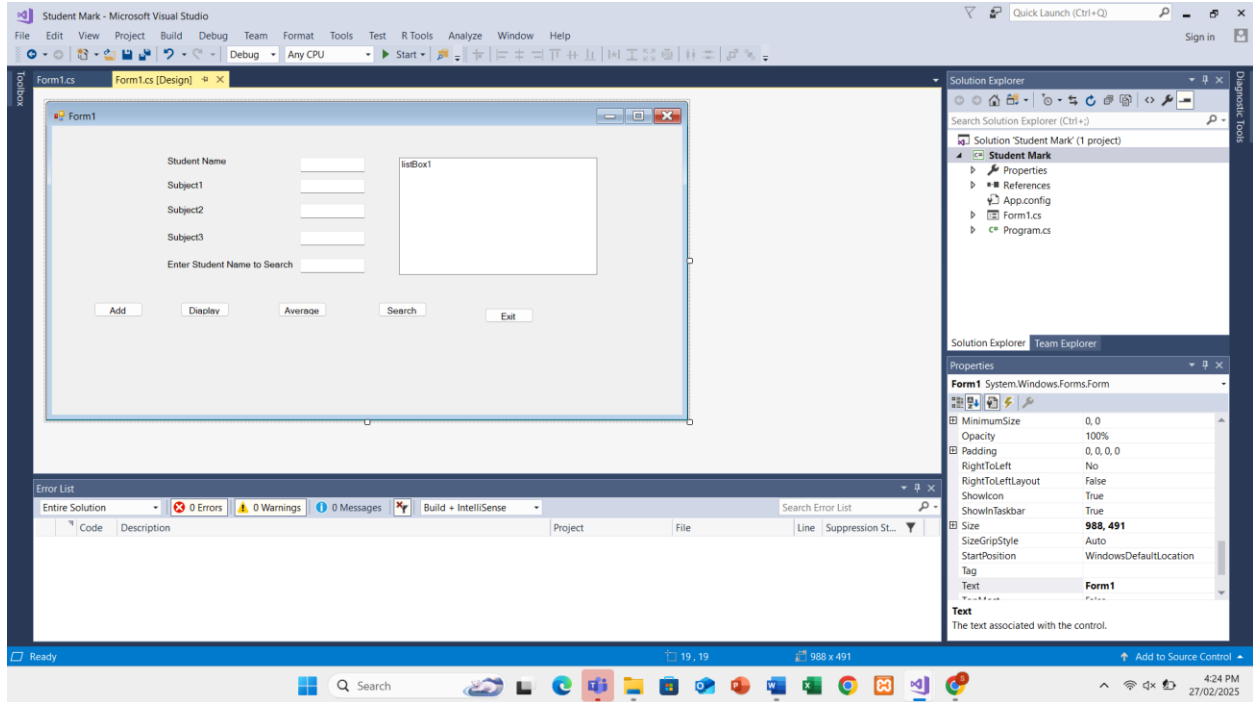
    private void Search_Click(object sender, EventArgs e)
    {
        string searchname = textBox5.Text;
        bool found = false;

        for (int i = 0; i < studentcount; i++)
        {
            if (StudentNames[i].Equals(searchname,
StringComparison.OrdinalIgnoreCase))
            {
                MessageBox.Show($" student found:{ StudentNames[i]}marks:{ StudentMarks[i,
0]},{ StudentMarks[i, 1]},{ StudentMarks[i, 2]}");
                found = true;
                break;
            }
        }
    }

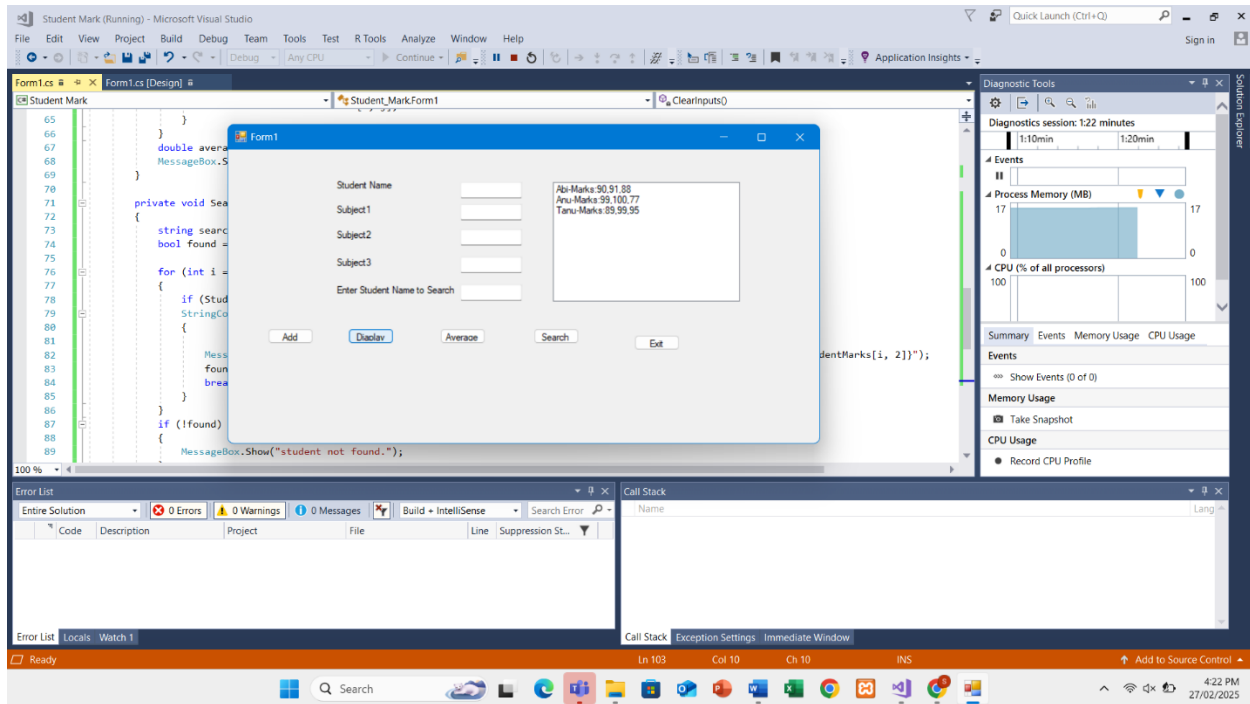
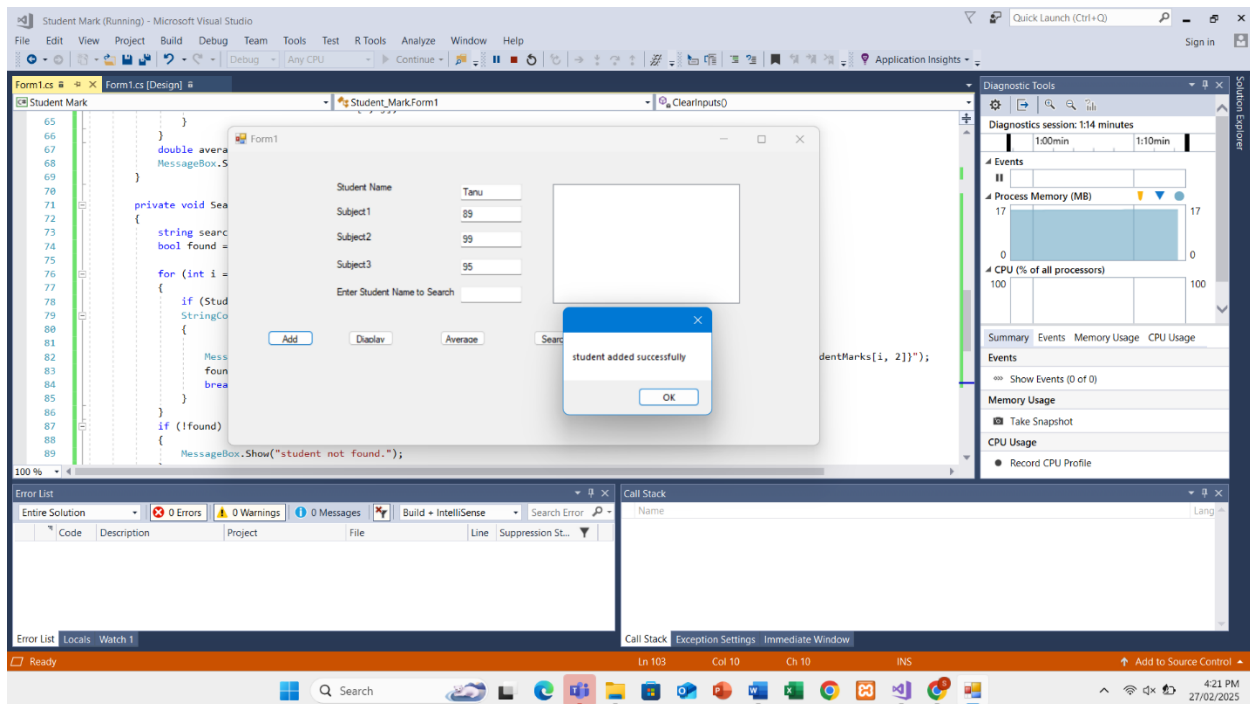
```

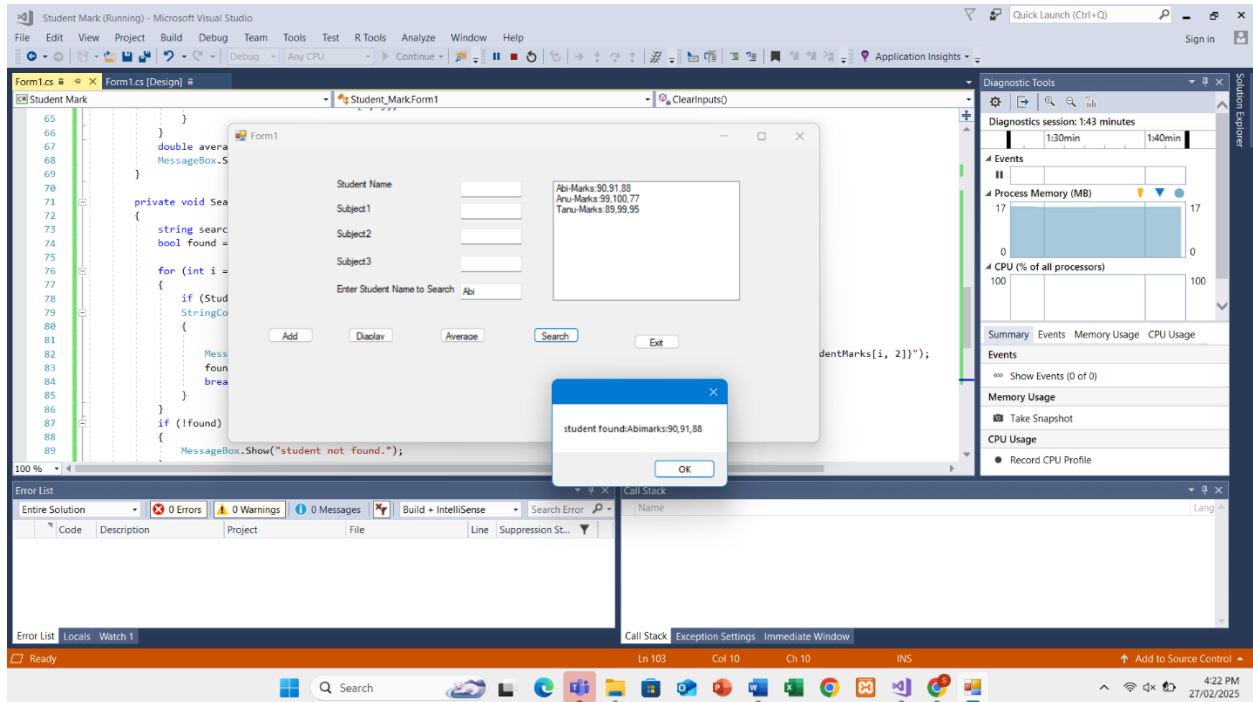
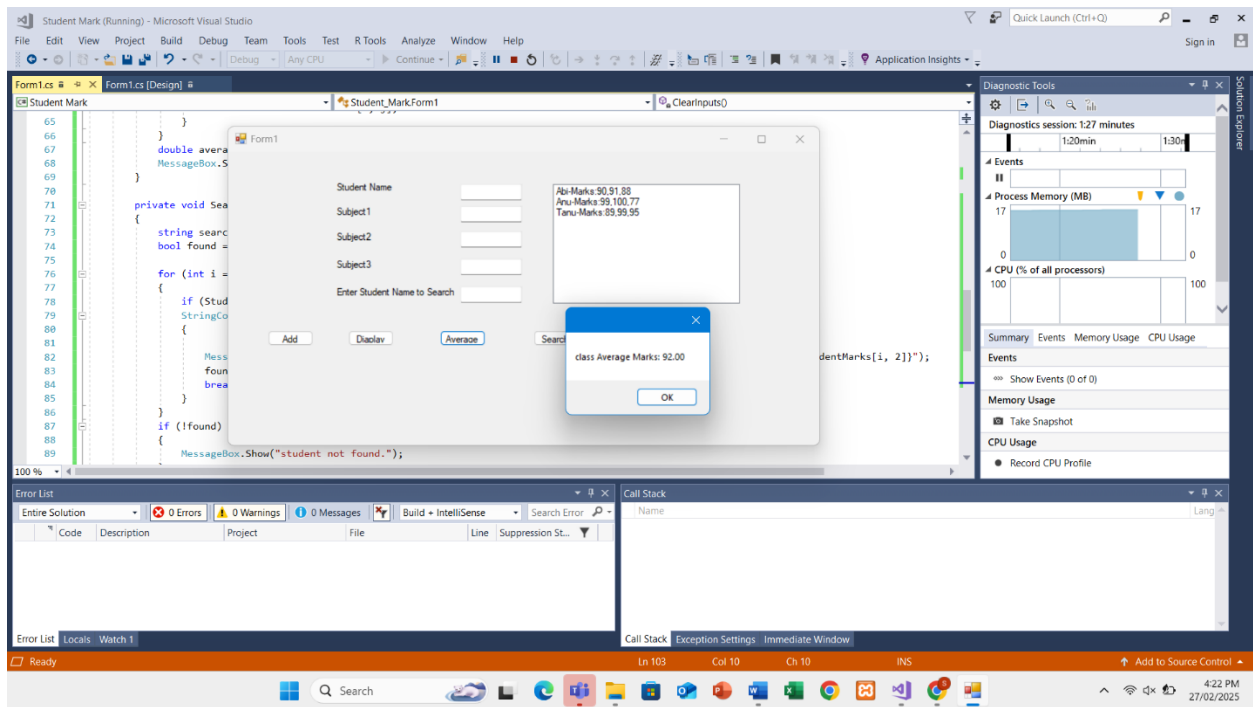
```
    }  
  }  
  if (!found)  
  {  
    MessageBox.Show("student not found.");  
  }  
}  
  
private void Exit_Click(object sender, EventArgs e)  
{  
  Environment.Exit(0);  
}  
private void ClearInputs()  
{  
  textBox1.Clear();  
  textBox2.Clear();  
  textBox3.Clear();  
  textBox4.Clear();  
}  
}  
}
```

Design Page:



Output:





4. Basic control classes

Source Code:

```
using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Drawing;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
using System.Windows.Forms;

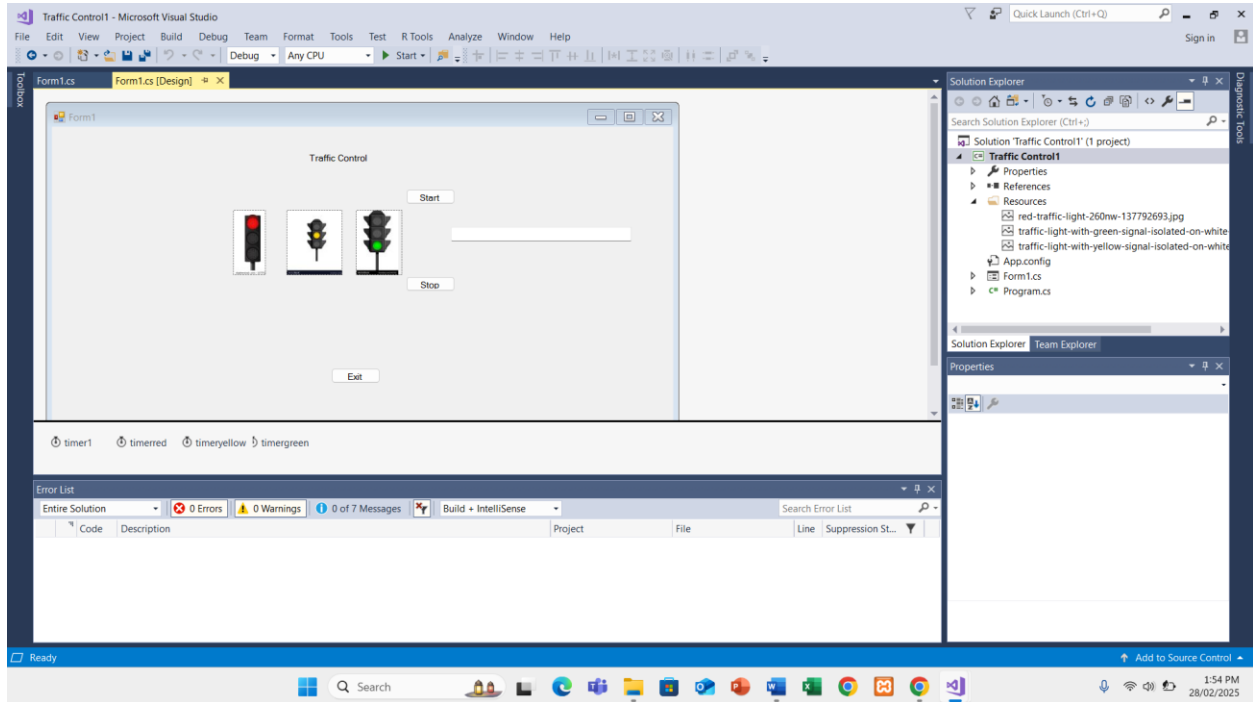
namespace Traffic_Control
{
    public partial class Form1 : Form
    {
        public Form1()
        {
            InitializeComponent();
            red.Visible = true;
            yellow.Visible = false;
            green.Visible = false;
        }
        private void timer1_Tick(object sender, EventArgs e)
        {
            if (red.Visible == true)
            {
                red.Visible = false;
                yellow.Visible = false;
                green.Visible = true;
                textBox1.Text = "Have a safe journey";
                timerred.Enabled = true;
            }
            else if (green.Visible == true)
            {
                red.Visible = false;
                yellow.Visible = true;
                green.Visible = false;
                textBox1.Text = "slow down your vehicle";
                timeryellow.Enabled = true;
            }
            else if (yellow.Visible == true){
                red.Visible = true;
            }
        }
    }
}
```

```

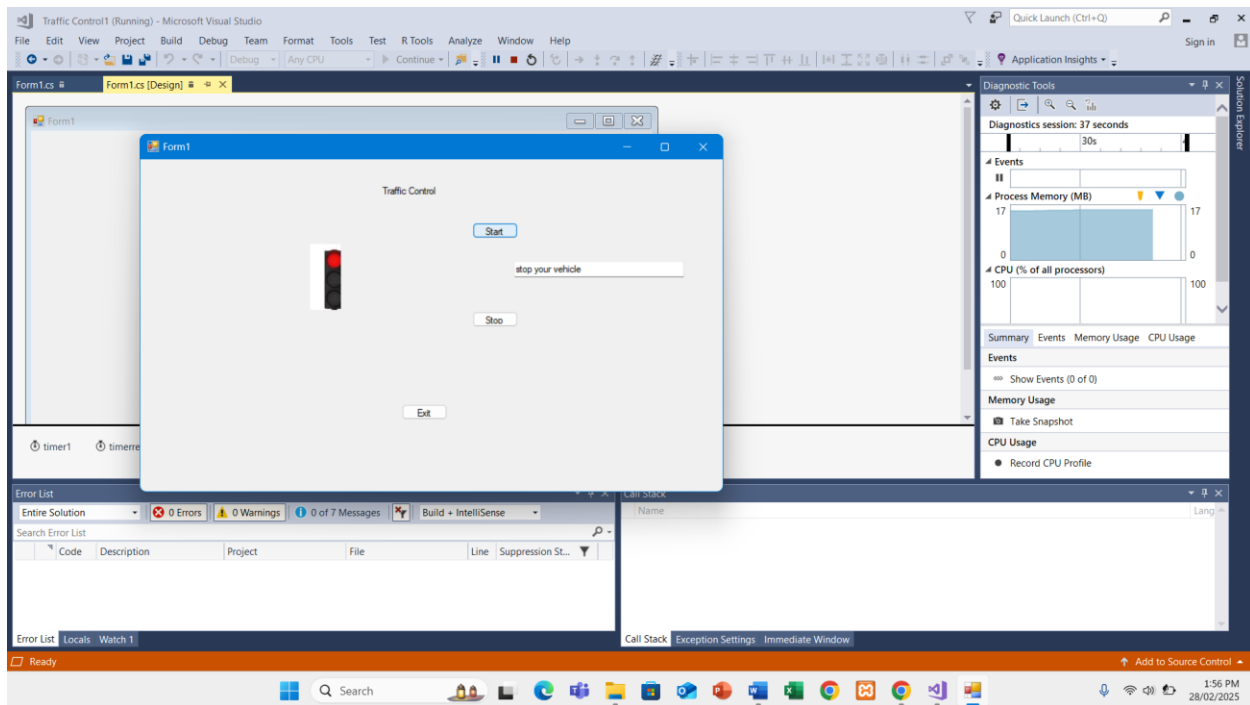
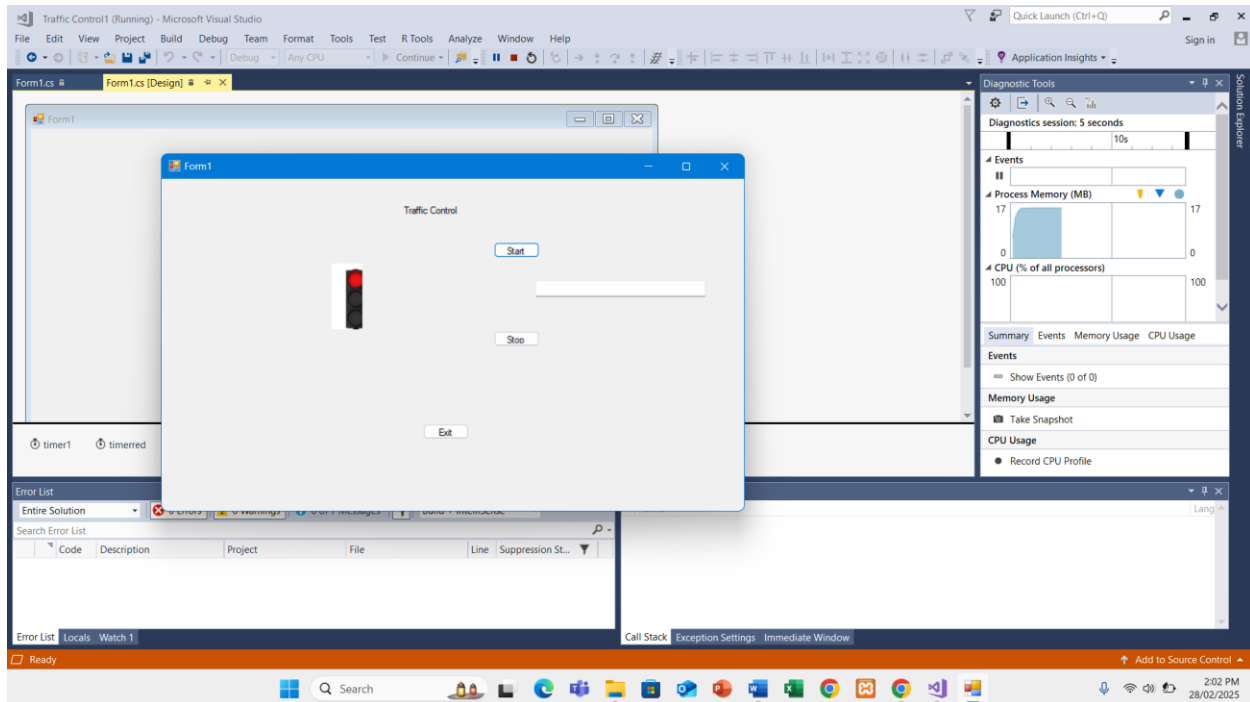
        yellow.Visible = false;
        green.Visible = false;
        textBox1.Text = "stop your vehicle";
        timergreen.Enabled = true;
    }
}
private void start_Click(object sender, EventArgs e)
{
    timerred.Enabled = true;
}
private void stop_Click(object sender, EventArgs e)
{
    timerred.Enabled = false;
    timeryellow.Enabled = false;
    timergreen.Enabled = false;
    red.Visible = true;
    yellow.Visible = false;
    green.Visible = false;
    textBox1.Text = "Stop your vehicle";
}
private void timerred_Tick(object sender, EventArgs e)
{
    timerred.Enabled = false;
    timer1_Tick(sender, e);
}
private void timeryellow_Tick(object sender, EventArgs e)
{
    timeryellow.Enabled = false;
    timer1_Tick(sender, e);
}
private void timergreen_Tick(object sender, EventArgs e)
{
    timergreen.Enabled = false;
    timer1_Tick(sender, e);
}
private void button3_Click(object sender, EventArgs e)
{
    Environment.Exit(0);
}
private void ClearInputs()
{
    textBox1.Clear();
}
}
}

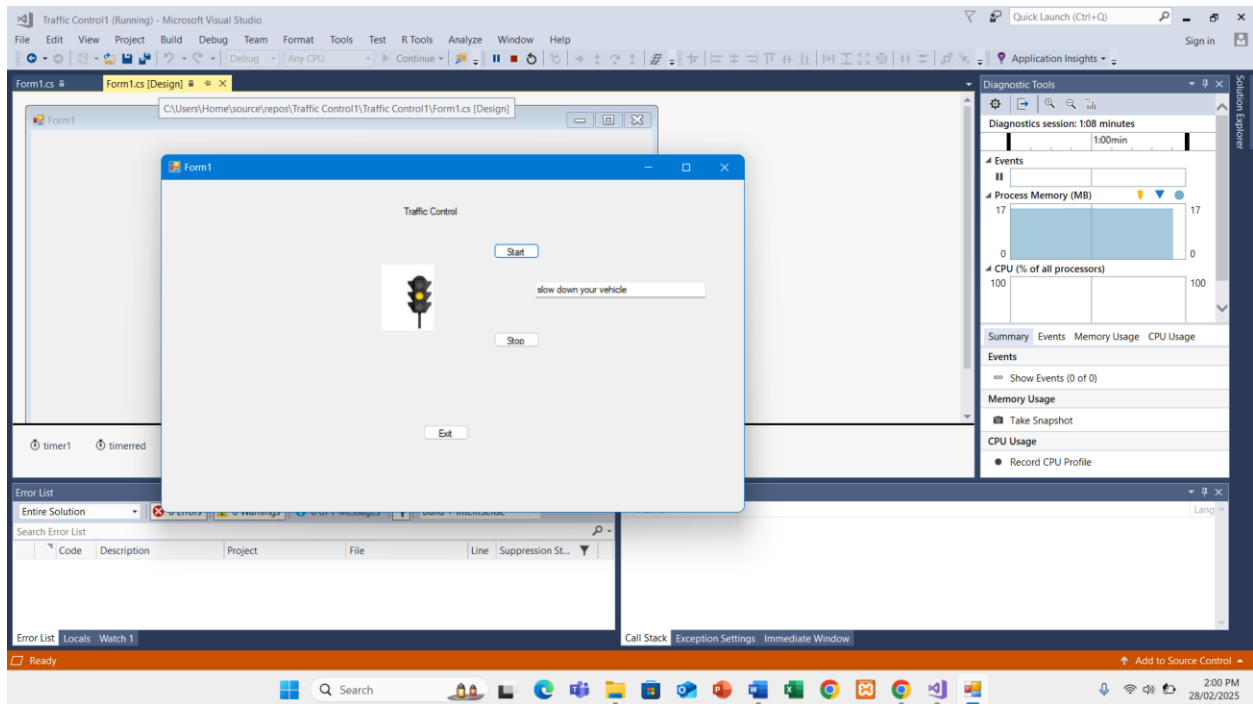
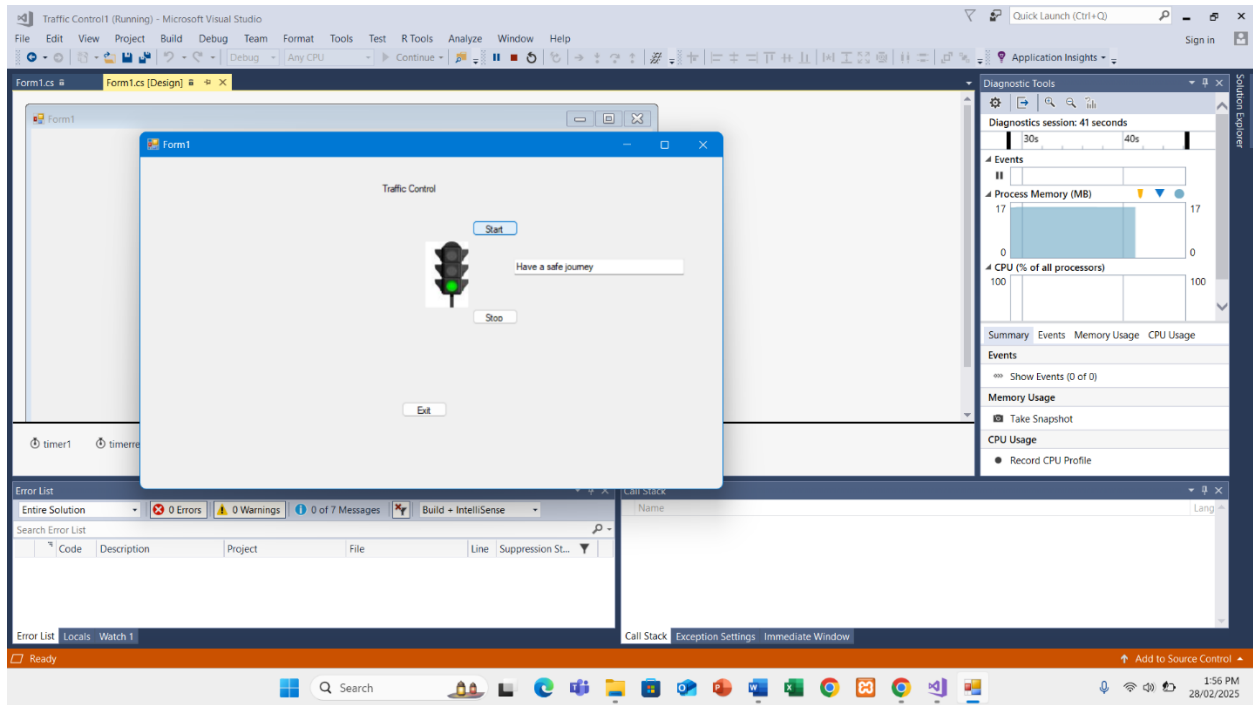
```

Design Page:



Output:





5. Exception Handling

Source Code (Form1):

```
using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Drawing;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
using System.Windows.Forms;

namespace Exception_Handling_1
{
    public partial class Form1 : Form{
        public Form1()
        {
            InitializeComponent();
        }
        private void button1_Click(object sender, EventArgs e){
            try{
                int numerator = Convert.ToInt32(textBox1.Text);
                int denominator = Convert.ToInt32(textBox2.Text);
                int Answer = numerator / denominator;
                textBox3.Text = Answer.ToString();
            }
            catch (FormatException){
                MessageBox.Show("Invalid input enter two numbers please");
            }
            catch (DivideByZeroException divideByZeroException){
                MessageBox.Show(divideByZeroException.Message, "Can't divide by zero");
            }
            finally{
                textBox1.Text = "";
                textBox2.Text = "";
            }
        }
        private void button2_Click(object sender, EventArgs e){
            var form2 = new Form2();
            form2.Show();
            this.Hide();
        }
    }
}
```

Form2:

```
using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Drawing;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
using System.Windows.Forms;

namespace Exception_Handling_1
{
    public partial class Form2 : Form
    {
        public Form2()
        {
            InitializeComponent();
        }

        private void button1_Click(object sender, EventArgs e)
        {
            try
            {
                int day = Convert.ToInt32(textBox1.Text);
                int month = Convert.ToInt32(textBox2.Text);
                int year = Convert.ToInt32(textBox3.Text);

                DateTime accessionDate = new DateTime(year, month, day);

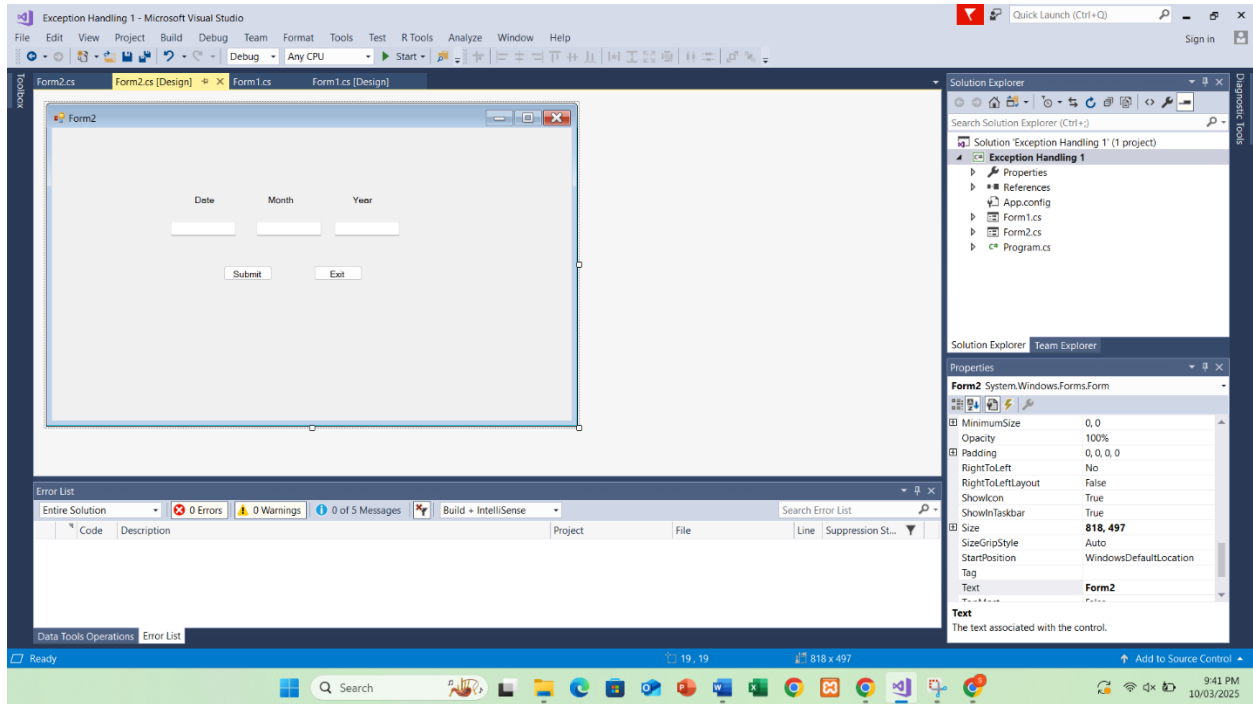
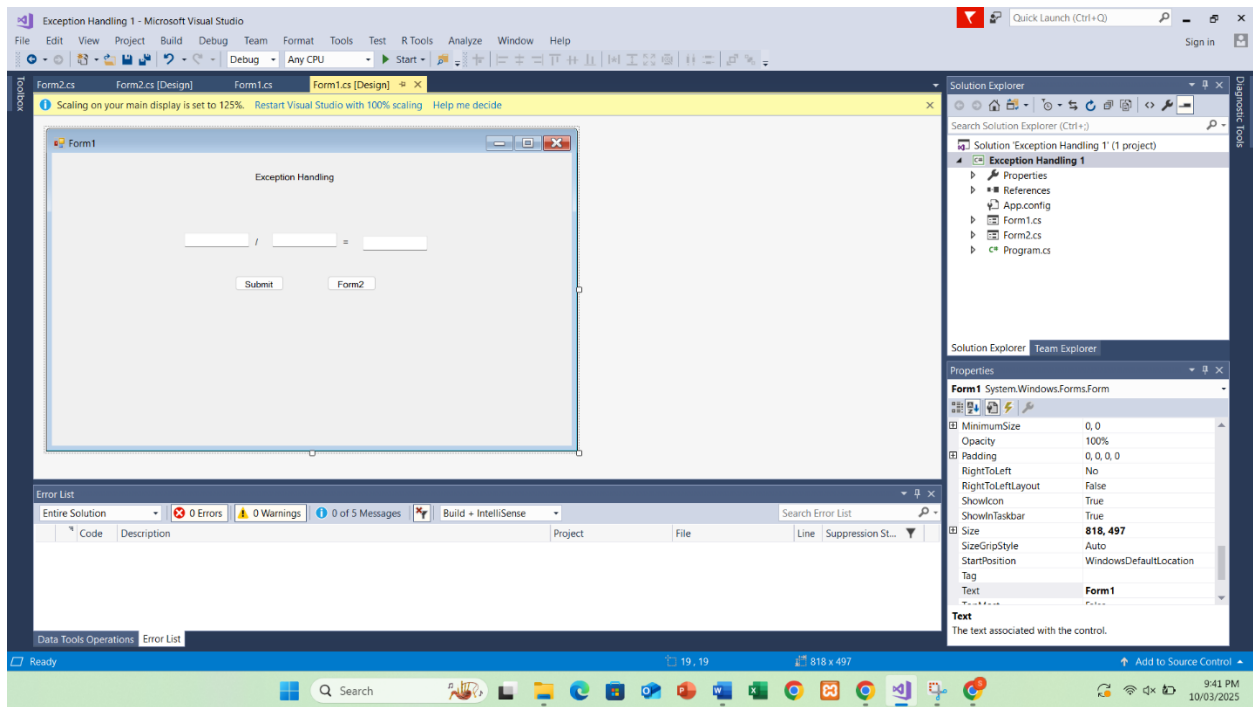
                MessageBox.Show($"Valid Accession Date:
{accessionDate.ToString("dd/MM/yyyy")}");
            }
            catch (FormatException)
            {
                MessageBox.Show("Invalid input. Please enter numeric values only.", "Input Error");
            }
            catch (ArgumentOutOfRangeException){
                MessageBox.Show("Invalid date. Please check the day, month, and year values.",
                "Date Error");
            }
            catch (Exception ex)
            {
                MessageBox.Show($"Unexpected error: {ex.Message}", "Error");
            }
        }
    }
}
```



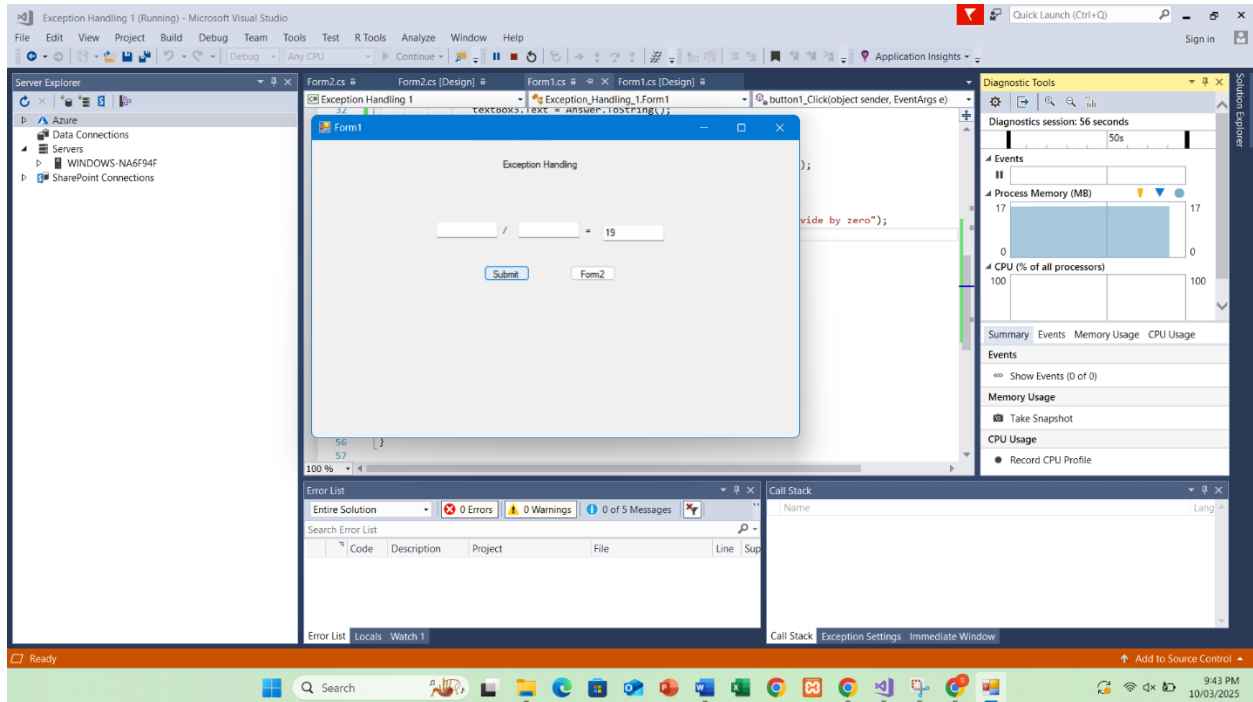
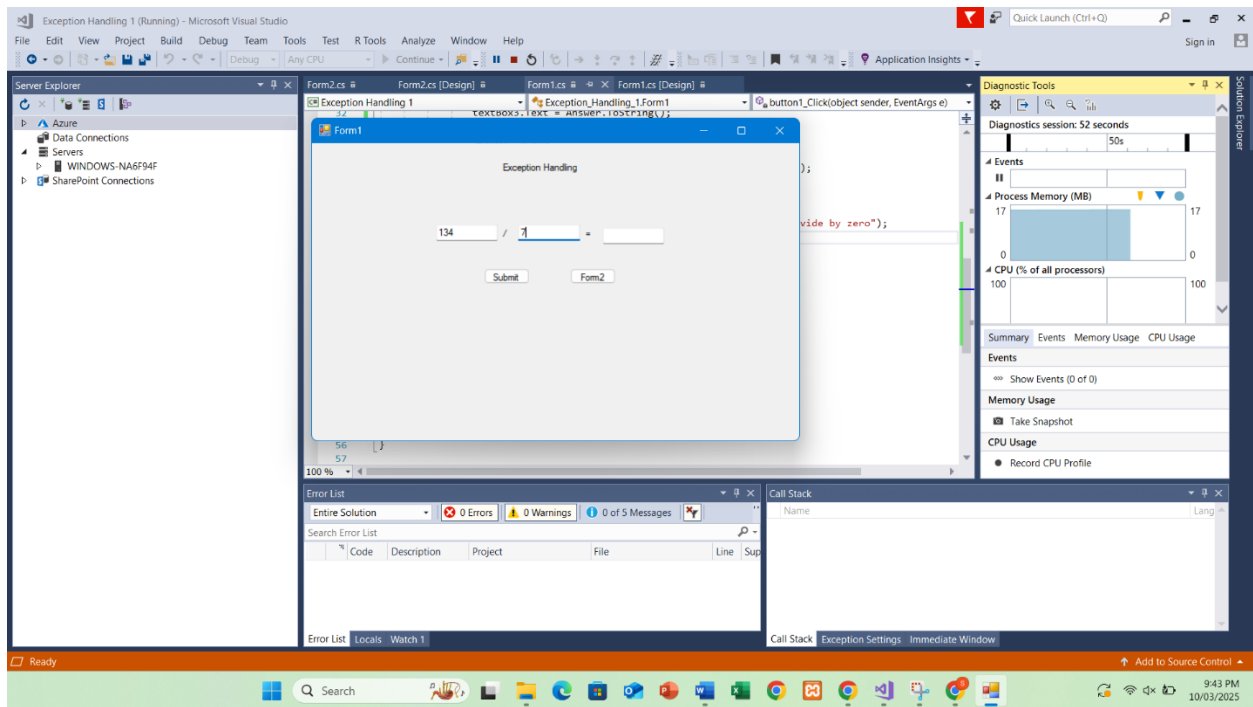
```
        Finally
        {
            textBox1.Clear();
            textBox2.Clear();
            textBox3.Clear();
        }
    }

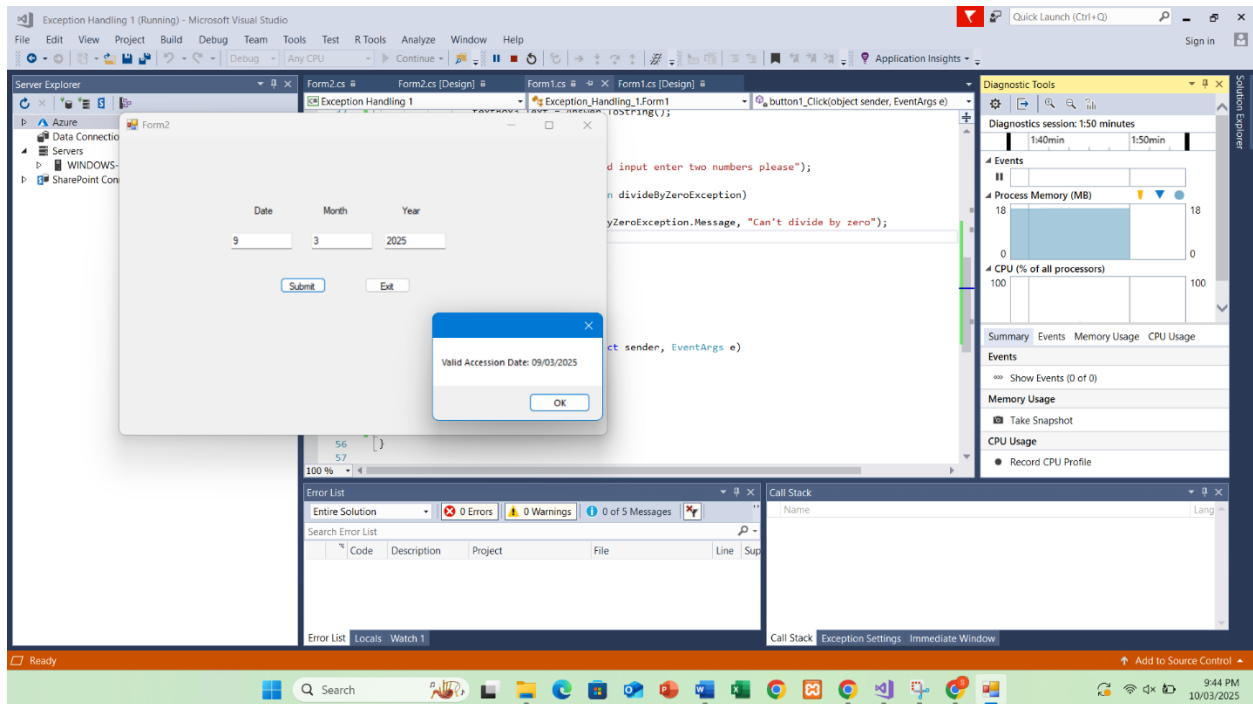
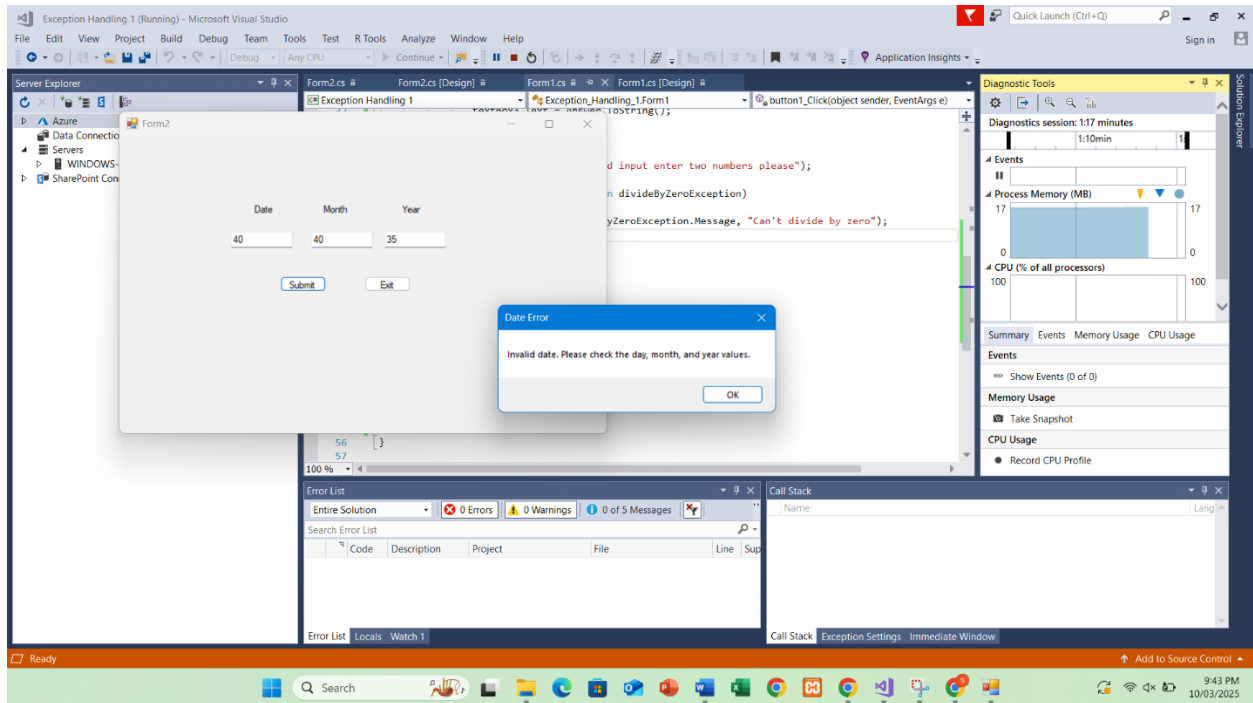
    private void button2_Click(object sender, EventArgs e)
    {
        Environment.Exit(0);
    }
}
```

Design Page:



Output:

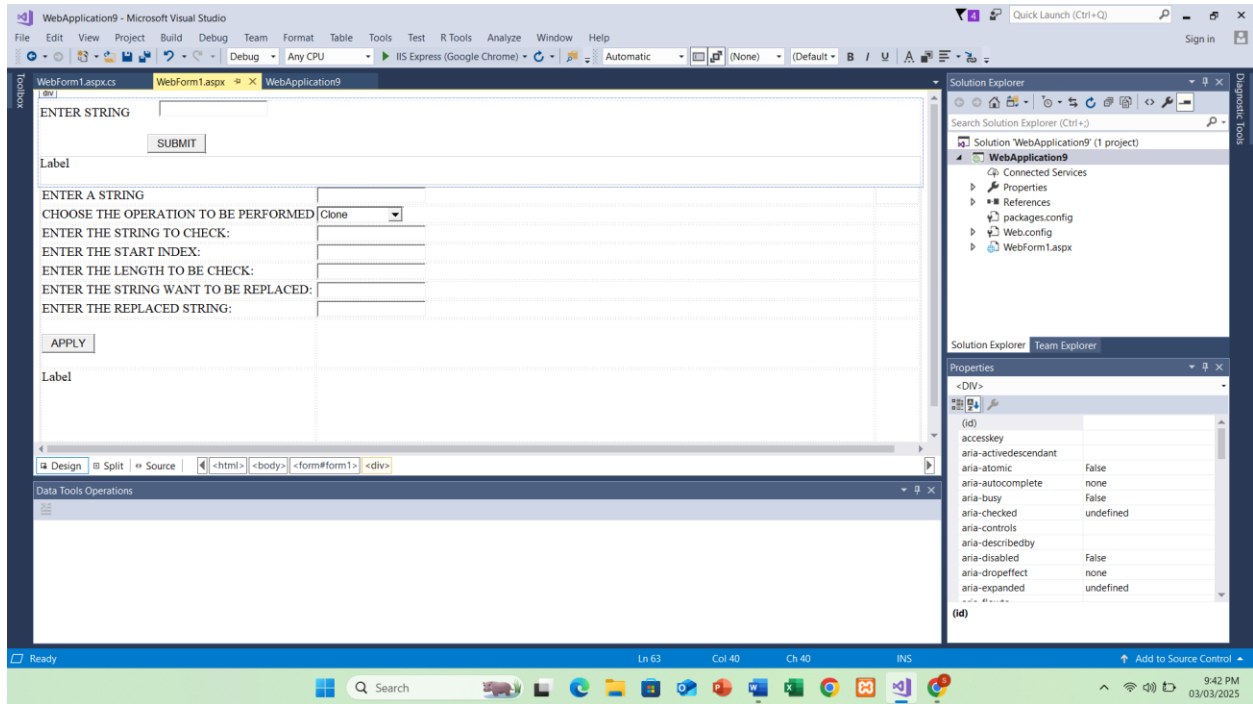





```
case 2:
    output = s.Contains(TextBox3.Text) ? "True" : "False";
    break;
case 3:
    output = s.ToLower();
    break;
case 4:
    output = s.ToUpper();
    break;
case 5:
    output = s.Trim();
    break;
case 6:
    int start = int.Parse(TextBox4.Text);
    int length = int.Parse(TextBox5.Text);
    output = s.Substring(start, length);
    break;
case 7:
    output = s.Replace(TextBox6.Text, TextBox7.Text);
    break;
default:
    output = "Invalid choice.";
    break;
}

Label11.Text = "Output: " + output;
}
}
```

Design Page:



Output:

localhost:49756/WebForm1.aspx X +

localhost:49756/WebForm1.aspx

ENTER STRING

number of digits:5
capital letters:1
small letters:9
spaces:2
special symbols:2

ENTER A STRING

CHOOSE THE OPERATION TO BE PERFORMED

ENTER THE STRING TO CHECK:

ENTER THE START INDEX:

ENTER THE LENGTH TO BE CHECK:

ENTER THE STRING WANT TO BE REPLACED:

ENTER THE REPLACED STRING:

Output: Hello world

Q Search 9:44 PM 03/03/2025

localhost:49756/WebForm1.aspx X +

localhost:49756/WebForm1.aspx

ENTER STRING

number of digits:5
capital letters:1
small letters:9
spaces:2
special symbols:2

ENTER A STRING

CHOOSE THE OPERATION TO BE PERFORMED

ENTER THE STRING TO CHECK:

ENTER THE START INDEX:

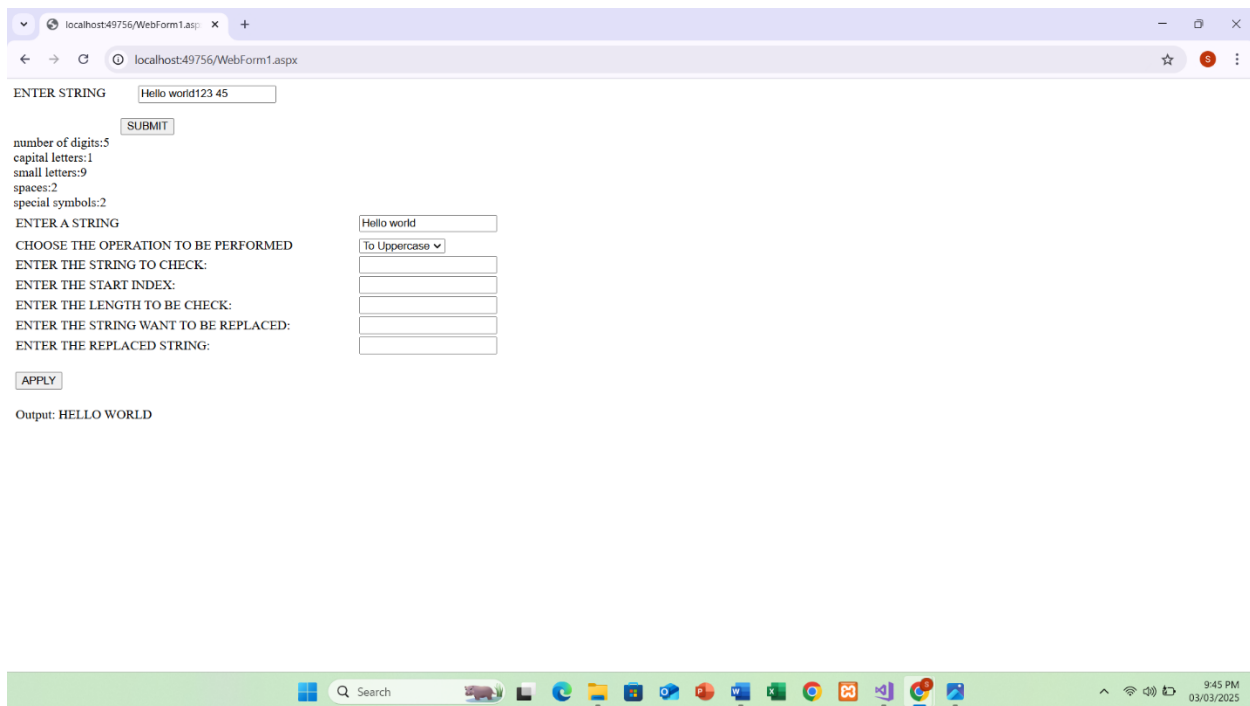
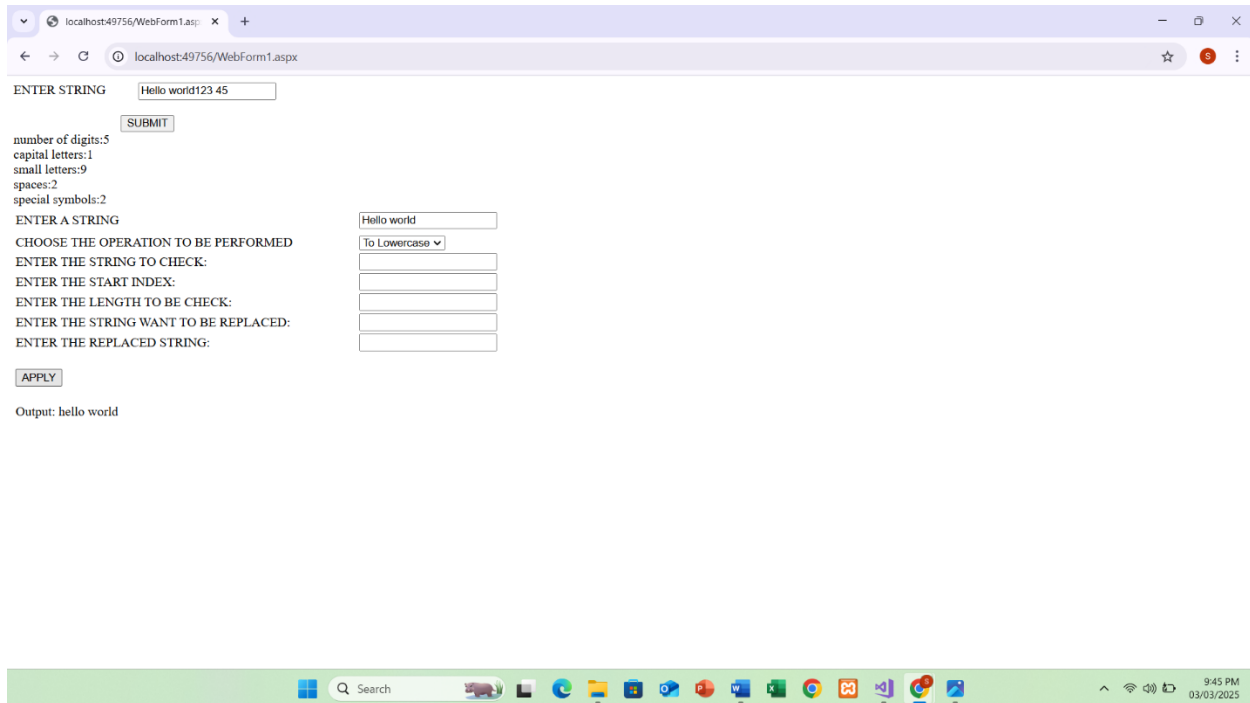
ENTER THE LENGTH TO BE CHECK:

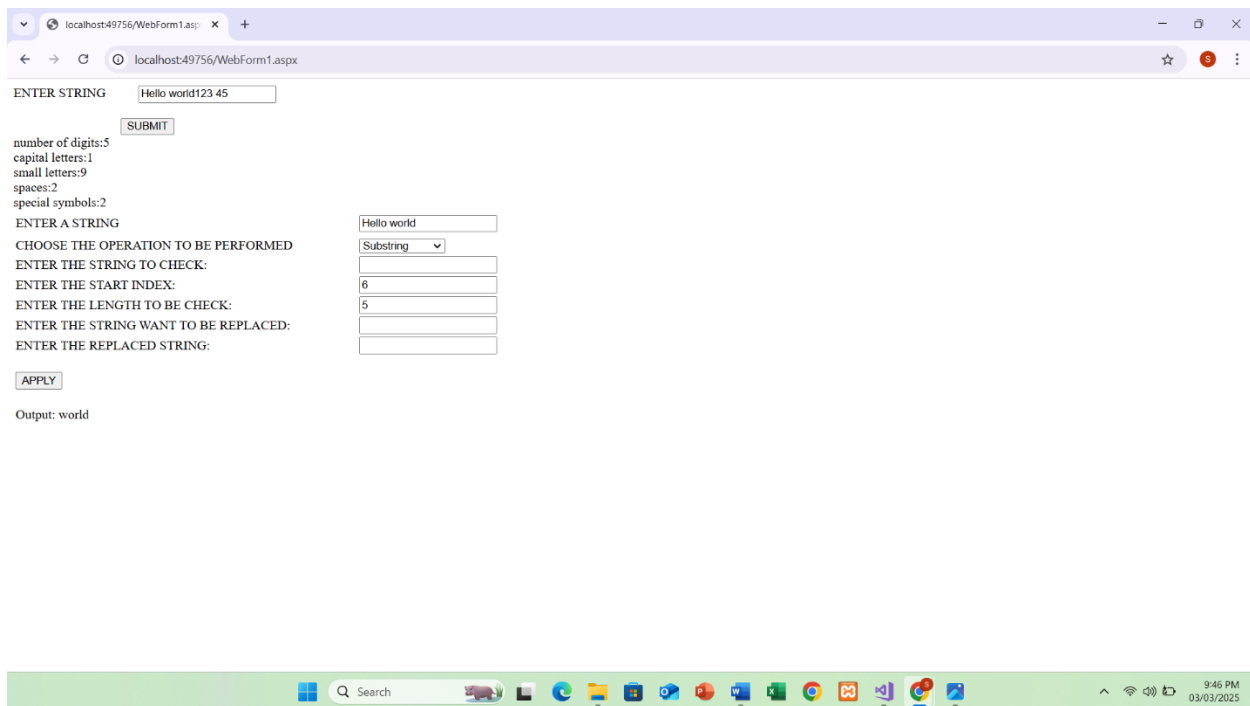
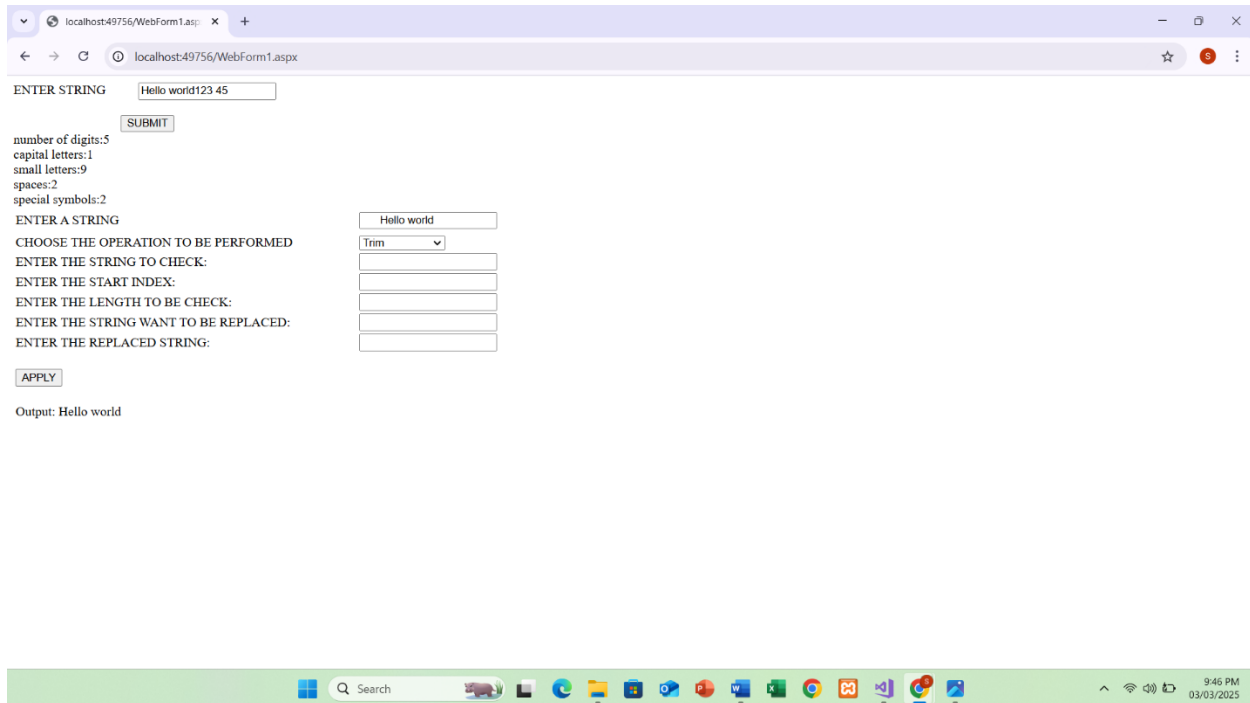
ENTER THE STRING WANT TO BE REPLACED:

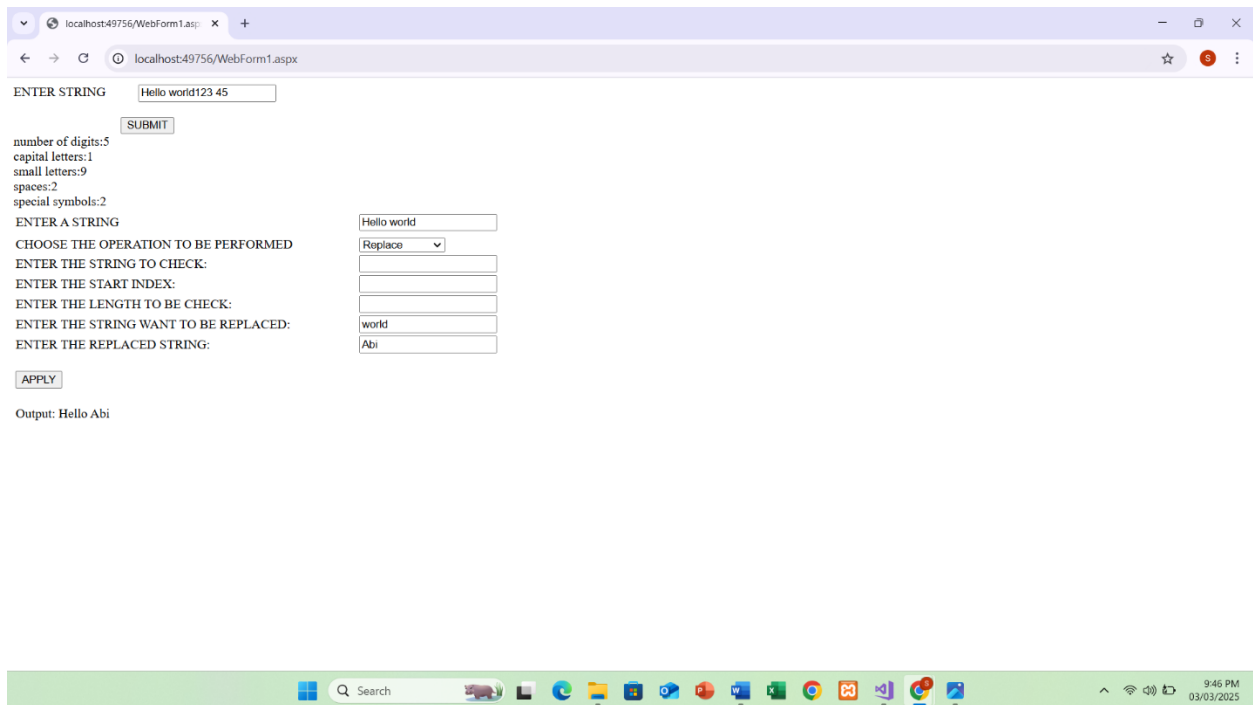
ENTER THE REPLACED STRING:

Output: False

Q Search 9:45 PM 03/03/2025







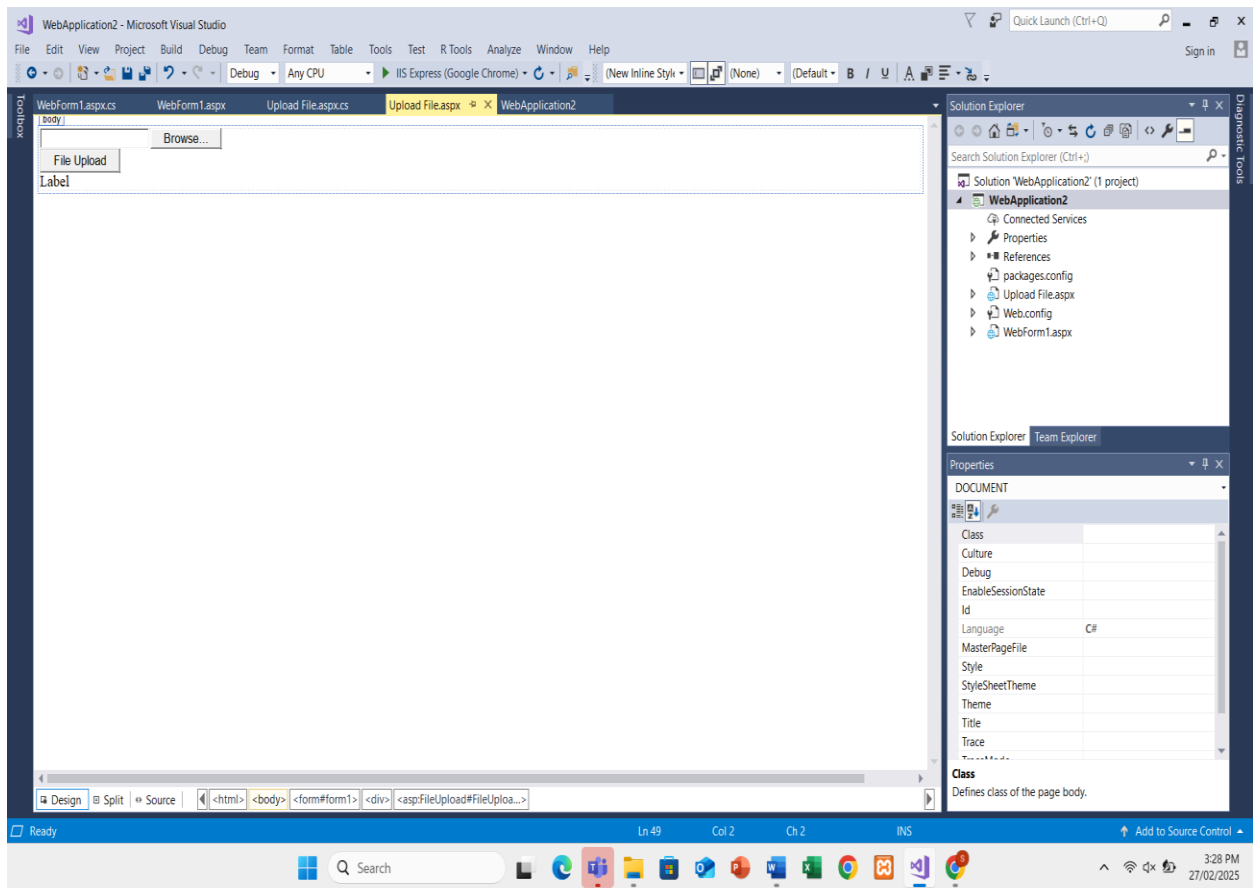
7. File operations

Source Code:

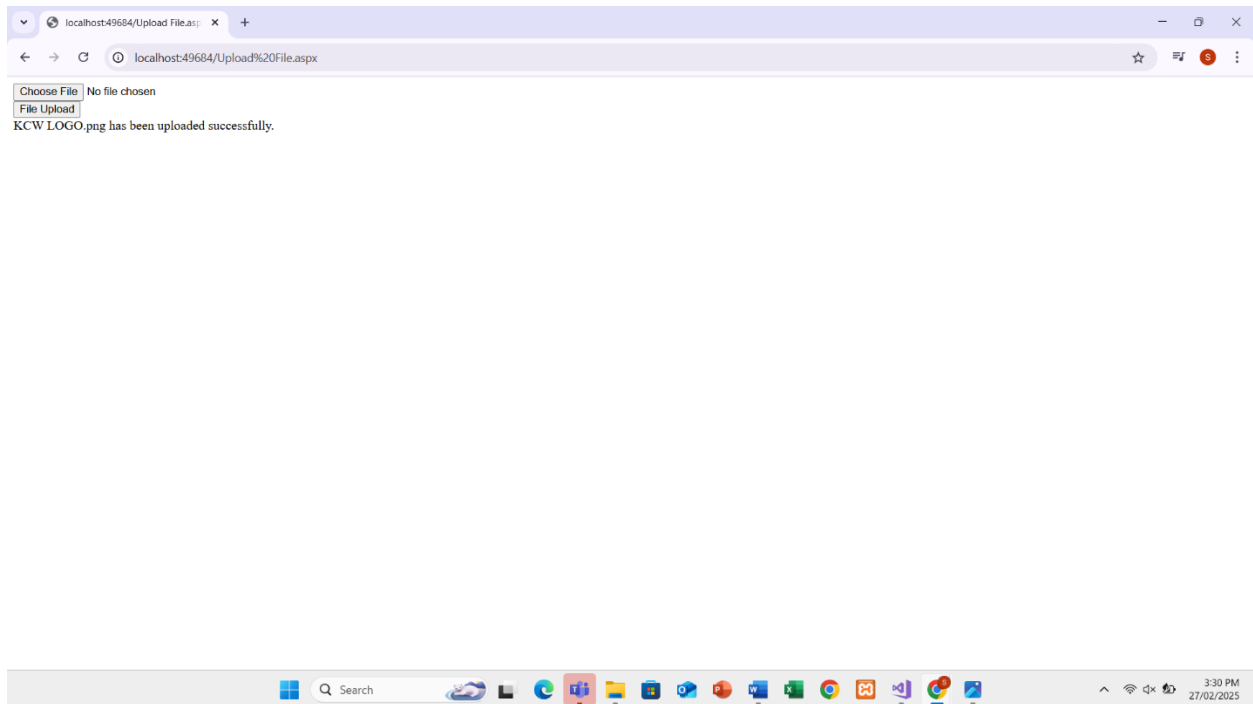
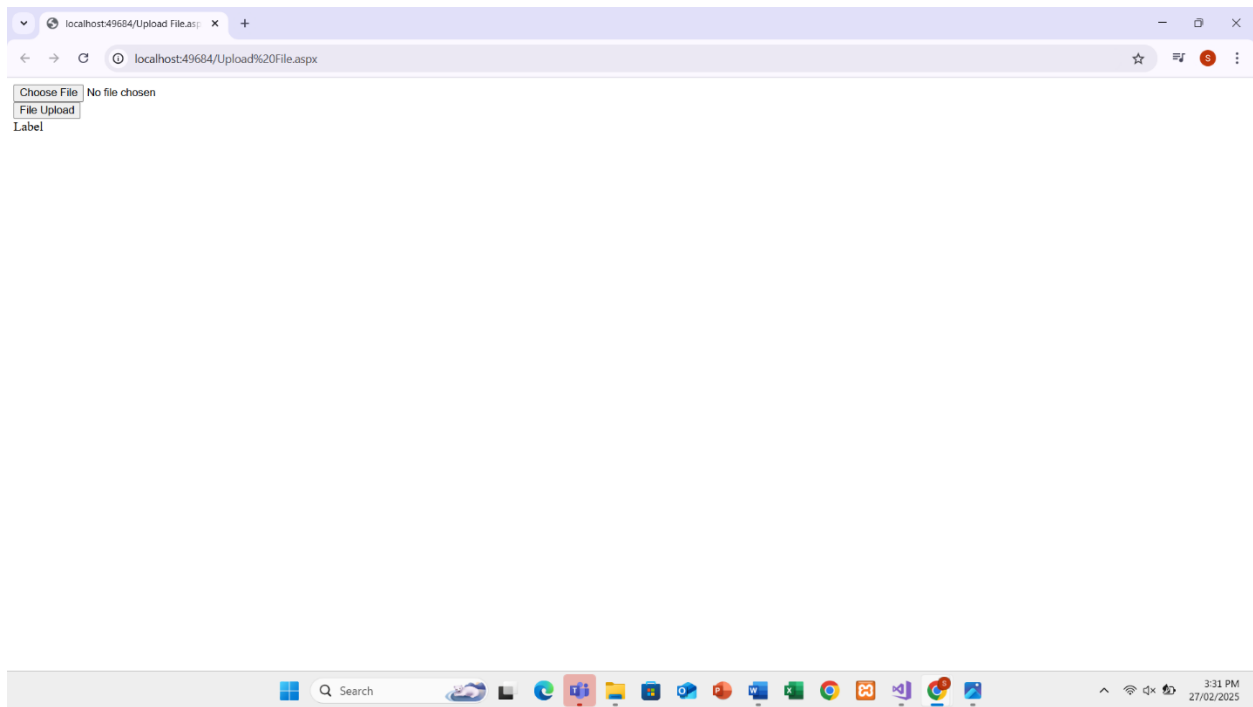
```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;
using System.IO;

namespace WebApplication2
{
    public partial class Upload_File : System.Web.UI.Page
    {
        protected void FileUpload1_DataBinding(object sender, EventArgs e)
        {
        }
        protected void Button1_Click(object sender, EventArgs e)
        {
            if (FileUpload1.HasFile)
            {
                string pathoffolder = Server.MapPath("~/upload images/");
                if (!Directory.Exists(pathoffolder))
                {
                    Directory.CreateDirectory(pathoffolder);
                }
                string getfilename = Path.GetFileName(FileUpload1.FileName);
                string savePath = Path.Combine(pathoffolder, getfilename);
                try
                {
                    FileUpload1.SaveAs(savePath);
                    Label1.Text = getfilename + " has been uploaded successfully.";
                }
                catch (Exception ex)
                {
                    Label1.Text = "Error: " + ex.Message;
                }
            }
            else
            {
                Label1.Text = "No file selected.";
            }
        }
    }
}
```

Design Page:



Output:



8. Master Pages, menu

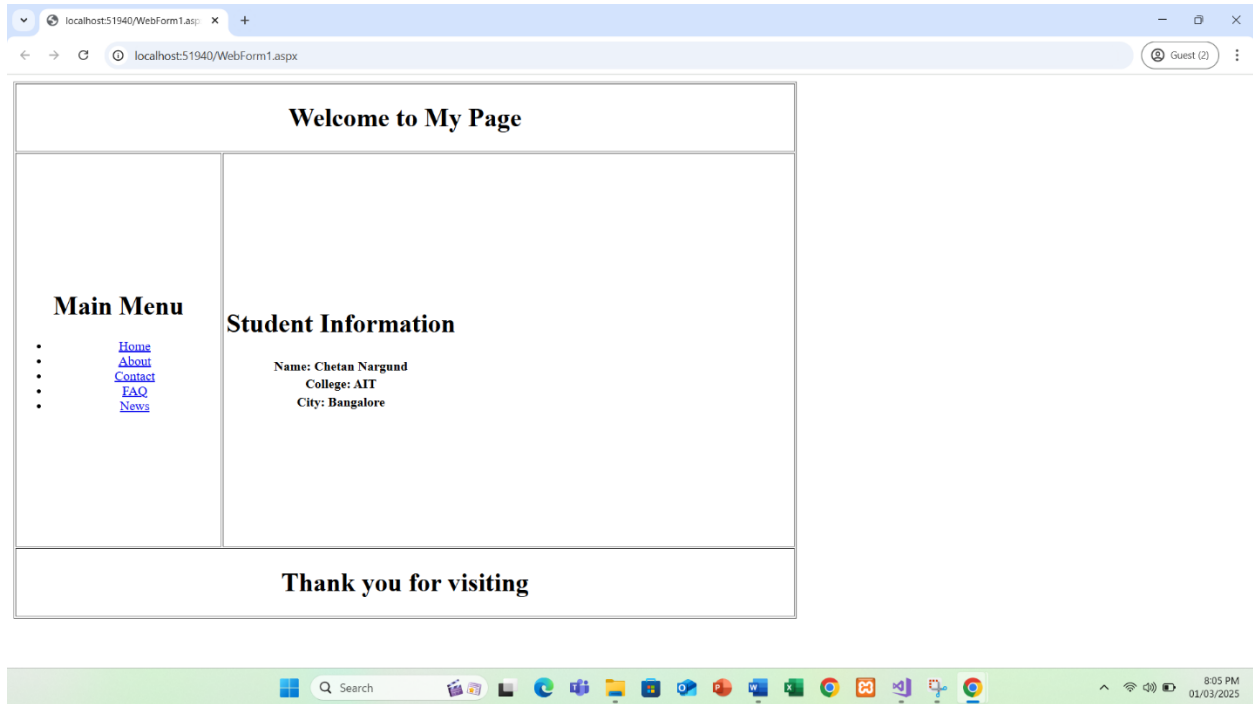
Source Code:

```
<%@ Master Language="C#" AutoEventWireup="true" CodeBehind="Site1.master.cs"
Inherits="Master_1.Site1" %>
<!DOCTYPE html>
<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
<title></title>
</head>
<body>
<form id="form2" runat="server">
<div>
<table border="1">
<tr>
<td colspan="2" style="text-align: center">
<h1>Welcome to My Page </h1>
</td>
</tr>
<tr>
<td style="text-align: center; height: 480px; width: 250px">
<h1>Main Menu</h1>
<ul>
<li style="margin-top: 0px"><a href="#">Home</a></li>
<li><a href="#">About</a></li>
<li><a href="#">Contact</a></li>
<li><a href="#">FAQ</a></li>
<li><a href="#">News</a></li>
</ul>
</td>
<td style="text-align: center; height: 480px; width: 700px">
<asp:ContentPlaceHolder ID="MainContentPlaceHolder1" runat="server">
<h1>You can change content here</h1>
</asp:ContentPlaceHolder>
</td>
</tr>
<tr>
<td colspan="2" style="text-align: center">
<h1>Thank you for visiting</h1>
</td>
</tr>
</table>
</div>
</form>
</body>
</html>
```

Content Page:

```
<%@ Page Title="" Language="C#" MasterPageFile="~/Site1.Master"
AutoEventWireup="true" CodeBehind="WebForm1.aspx.cs" Inherits="Master_1.WebForm1"
%>
<asp:Content ID="Content1" ContentPlaceHolderID="MainContentPlaceHolder1"
runat="Server">
<table>
<tr>
<th>
<h1>Student Information</h1>
</th>
</tr>
<tr>
<td><b>Name: Chetan Nargund</b></td>
</tr>
<tr>
<td><b>College: AIT</b></td>
</tr>
<tr>
<td><b>City: Bangalore</b></td>
</tr>
</table>
</asp:Content>
```


Output:



11. Content Linking

WebForm1.aspx:

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="WebForm1.aspx.cs"
Inherits="Content_Linking.WebForm1" %>

<!DOCTYPE html>
<html>
<body>
    <h1>This is Page 1!</h1>
    <asp:Literal ID="litNavigation" runat="server"></asp:Literal>
    <p>The example uses the Content Linking Component to navigate between the pages in a text
file.</p>
    <p>
        <a href="text/links2.txt"></a>
    </p>
</body>
</html>
```

WebForm2.aspx:

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="WebForm2.aspx.cs"
Inherits="Content_Linking.WebForm2" %>

<!DOCTYPE html>
<html>
<body>
    <h1>This is Page 2!</h1>
    <asp:Literal ID="litNavigation" runat="server"></asp:Literal>
</body>
</html>
```

WebForm3.aspx:

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="WebForm3.aspx.cs"
Inherits="Content_Linking.WebForm3" %>

<!DOCTYPE html>
<html>
<body>
    <h1>This is Page 3!</h1>
    <asp:Literal ID="litNavigation" runat="server"></asp:Literal>
</body>
</html>
```

WebForm1.aspx.cs, WebForm2.aspx.cs, WebForm3.aspx.cs (Same for All)

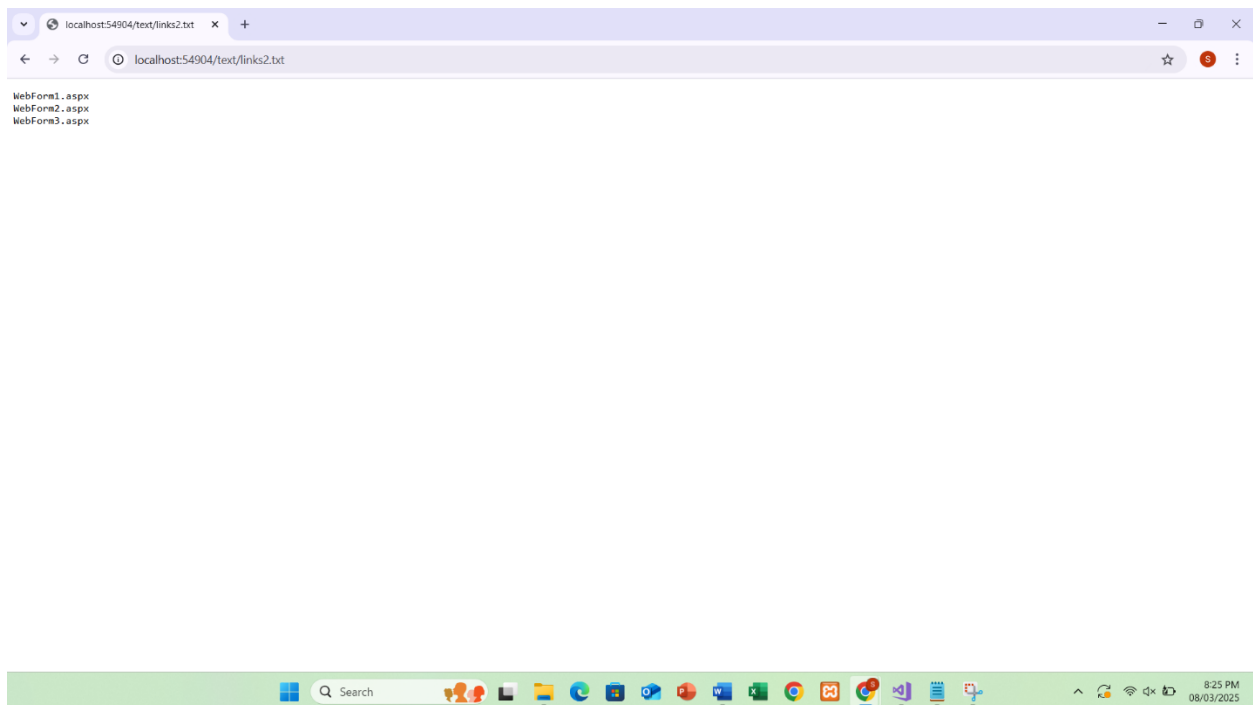
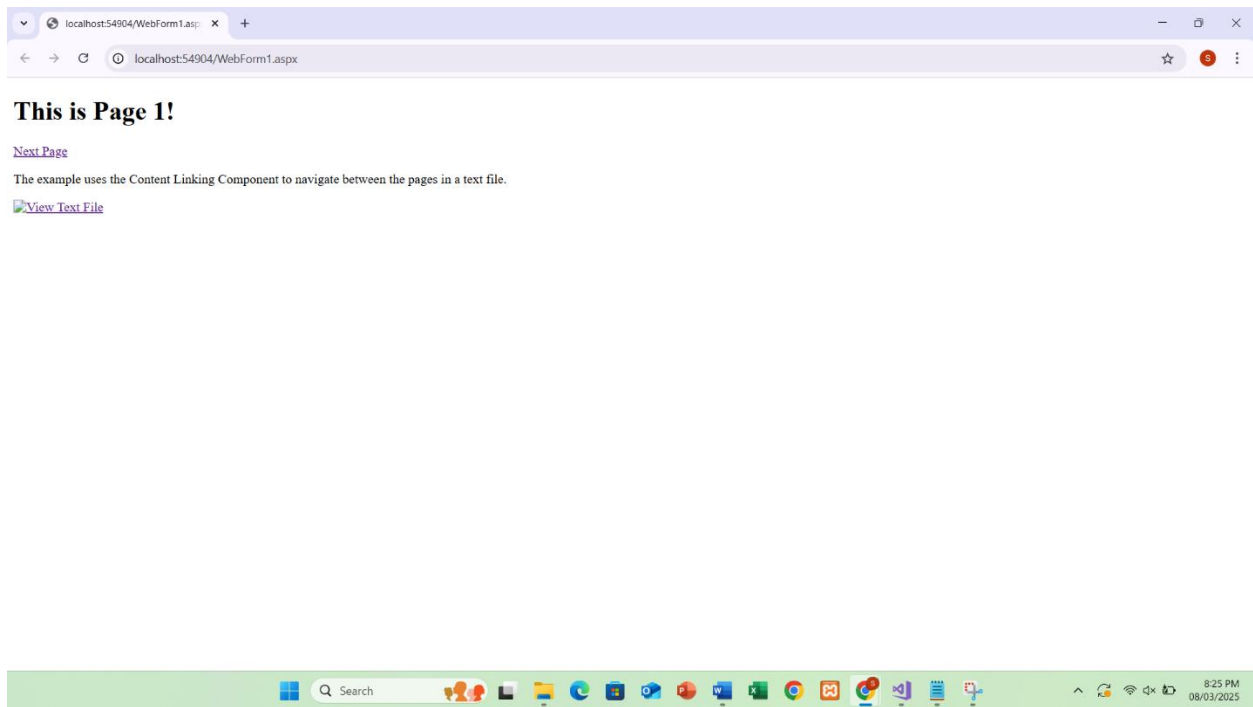
```
using System;
using System.IO;
using System.Web.UI;

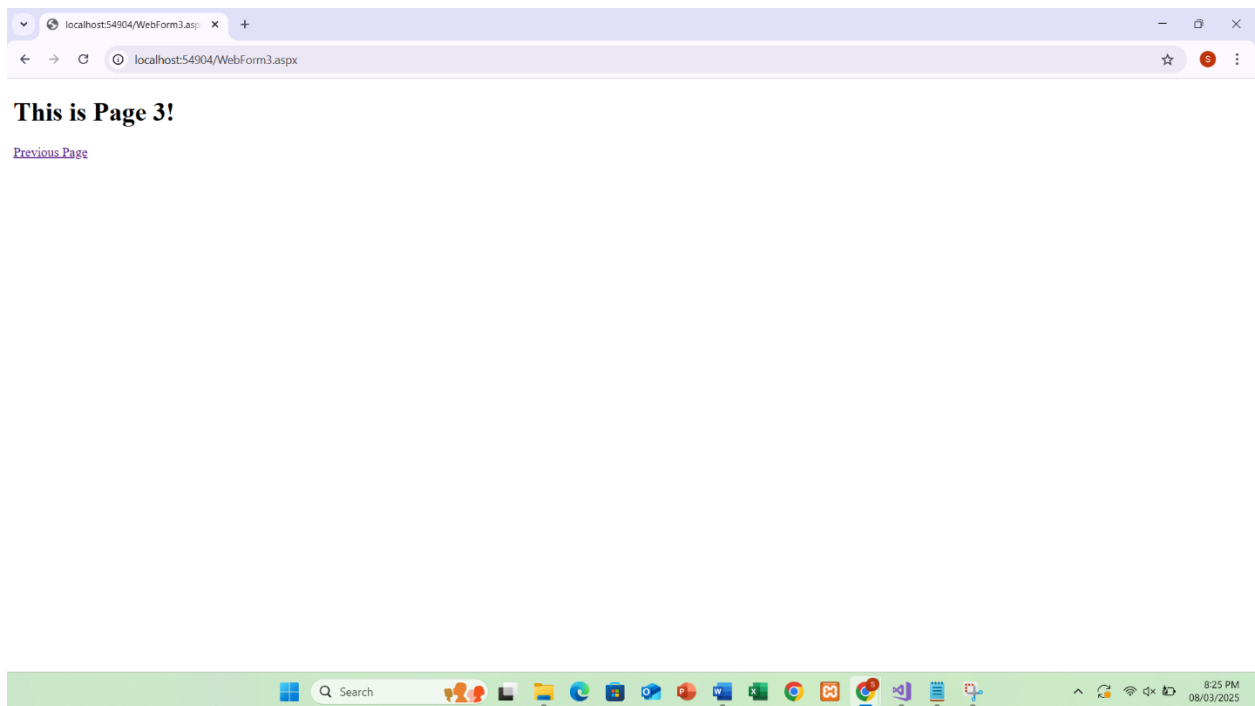
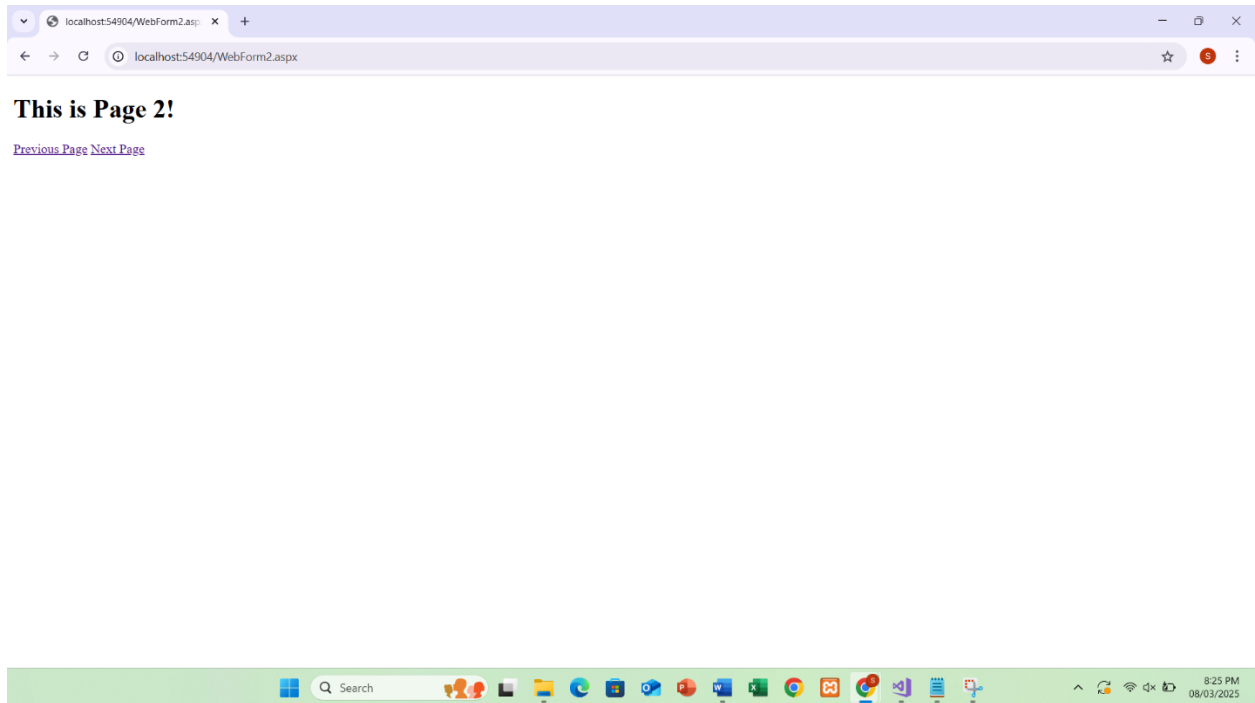
namespace Content_Linking
{
    public partial class WebForm1 : Page
    {
        protected void Page_Load(object sender, EventArgs e)
        {
            string path = Server.MapPath("~/text/links2.txt");
            if (!File.Exists(path)) return;

            string[] pages = File.ReadAllLines(path);
            int index = Array.IndexOf(pages, Path.GetFileName(Request.Url.AbsolutePath));

            litNavigation.Text = (index > 0 ? $"<a href='{pages[index - 1]}'>Previous Page</a> " :
            "") + (index < pages.Length - 1 ? $"<a href='{pages[index + 1]}'>Next Page</a>" : "");
        }
    }
}
```

Output:





12. Content Rotator

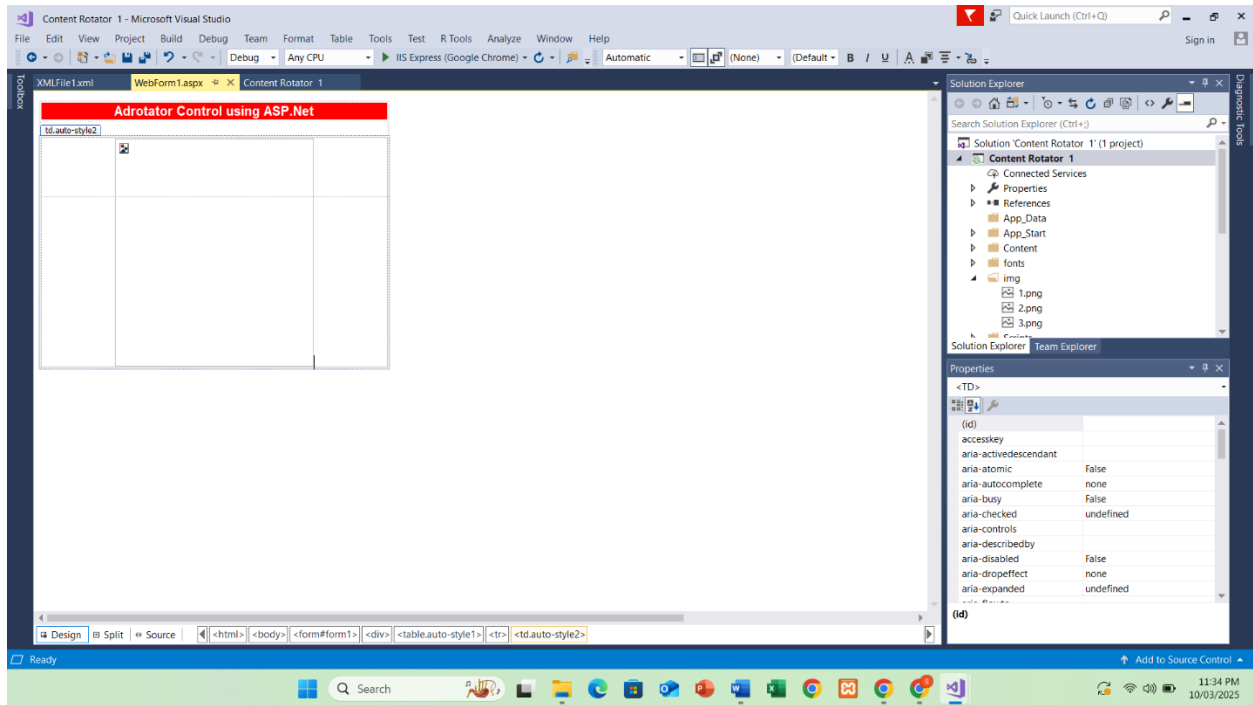
Source Code:

```
<?xml version="1.0" encoding="utf-8" ?>
<Advertisements>
  <Ad>
    <ImageUrl>img/1.png</ImageUrl>
    <NavigateUrl>https://meeraacademy.com</NavigateUrl>
    <AlternateText>Meera Academy</AlternateText>
    <Impressions>50</Impressions>
    <Keyword>Meera</Keyword>
  </Ad>

  <Ad>
    <ImageUrl>img/2.png</ImageUrl>
    <NavigateUrl>http://meera.com</NavigateUrl>
    <AlternateText>Meerta Aademy</AlternateText>
    <Impressions>100</Impressions>
    <Keyword>Academy</Keyword>
  </Ad>

  <Ad>
    <ImageUrl>img/3.png</ImageUrl>
    <NavigateUrl>https://meeraacademy.com</NavigateUrl>
    <AlternateText>Meerta Aademy</AlternateText>
    <Impressions>50</Impressions>
    <Keyword>Academy</Keyword>
  </Ad>
</Advertisements>
```

Design Page:



Output:

